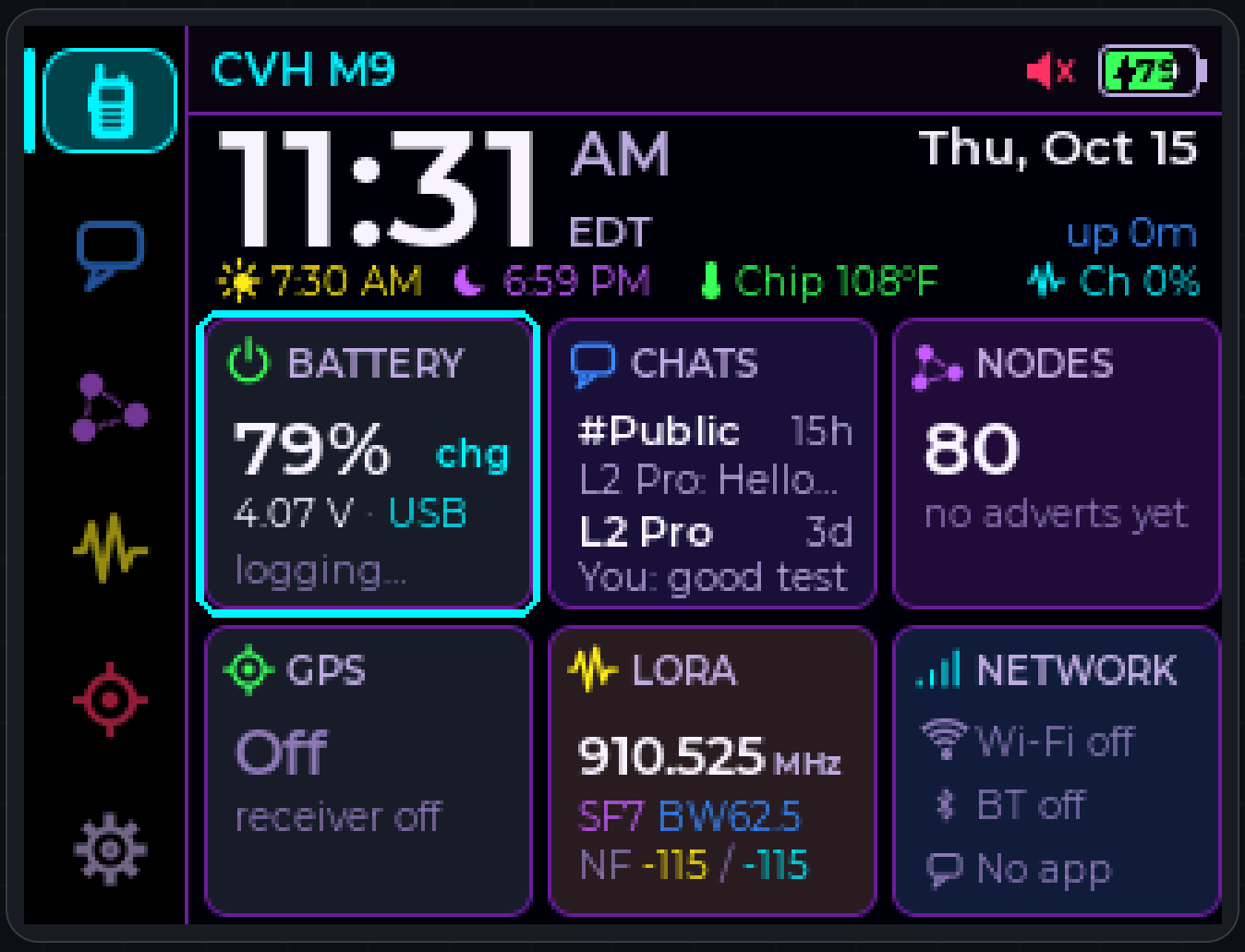
CAIRN.

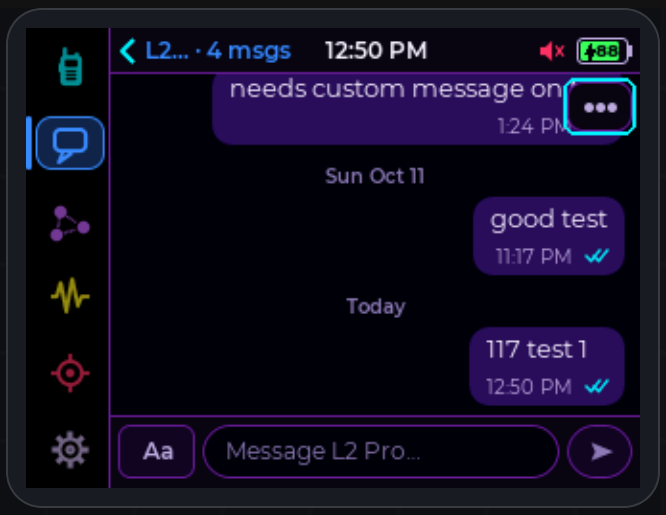
Firmware manual.

Cairn is a MeshCore firmware with its own interface on the device. Read and send messages, find nodes, watch the radio, follow the map and change every setting from the screen, with or without a phone.

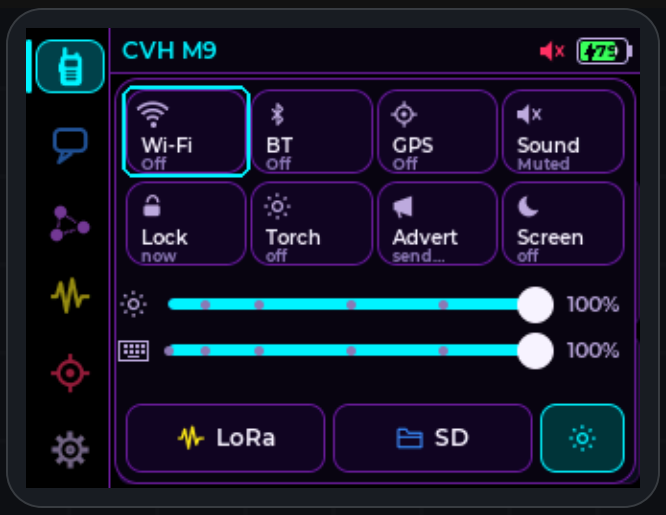
CAIRN 118 · OCTOBER 2026 | TOUCH · KEYS · OLED | BY CVH VISUALS



M9 Home, with the key focus ring on the Battery card.



Delivered, two ticks.



The quick panel.

INTERFACES

4

COMPANION BUILDS

9

BASE

MeshCore 1.17.1

LICENCE

MIT

01 / ABOUT CAIRN

One firmware, many radios.

Cairn is a fork of the MeshCore companion firmware. Every board is built from one source tree that shares MeshCore's mesh and radio code, so every build still works with the official MeshCore apps on Android, iOS and the web.

What Cairn adds is the device itself: a screen you can run the mesh from. On the larger boards that is a full interface with a sidebar, cards and pages. On the small OLED and e-ink boards it is a compact page-by-page interface. Each build is numbered. The boot splash shows **BUILD N**, releases are tagged `cairn-vN`, and files are named `MeshCore-<Board>-CairnN-<date>…`, so you can match a device to a release from its own screen.

Four interfaces

Cairn touch and key interface

Sidebar, Home dashboard, chats, nodes, radios, GPS with offline maps, settings as cards. Driven by touch, by keys, or both.

L2 PRO · HELTEC V4 KIT · THINKNODE M9 · T-DECK

T-Display SF32 interface

A native card interface for the 480×480 AMOLED, run from a 20-key phone keypad or touch.

LILYGO T-DISPLAY SF32

Compact interface

Page by page on small OLED, e-ink and colour TFT screens, run from a joystick or one button.

WIO L1 PRO · DV1 · L1 E-INK · T096

Compact repeater

The standard MeshCore repeater with a minimal status, spectrum and settings screen.

T096 · T114 · HELTEC V2-V4 · TRACKERS

Supported boards

BOARD	SCREEN AND INPUT	INTERFACE	STATUS
Seeed Wio Tracker L2 Pro	3.2" 320×240 touchscreen	Cairn touch	TESTED
Heltec V4-R8 + Expansion Kit V2	2.8" touchscreen	Cairn touch	TESTED
Elecrow ThinkNode M9	2.4" screen, QWERTY keyboard, d-pad, function keys	Cairn keys	BETA
LilyGO T-Deck / T-Deck Plus	2.8" touchscreen, keyboard, trackball	Cairn touch and keys	BETA
LilyGO T-Display SF32 + keypad	480×480 AMOLED touch, 20-key keypad	SF32	TESTED
Seeed Wio Tracker L1 Pro (OLED, DV1, e-ink)	SH1106 OLED, 2.42" SSD1309 OLED or 2.13" e-ink; joystick	Compact	TESTED
Heltec Mesh Node T096	0.96" colour TFT, one button	Compact	TESTED
Compact repeaters	T096, T114, Heltec V2, V3, V4, V4-R8, Wireless Tracker V1/V2	Repeater	T096 BETA OTHERS UNTRIED

About 30 more boards build from the same tree and sit in a separate, untested preview release (`cairn-v107-preview`). This manual covers the boards above.

How to read this manual

Most chapters describe the Cairn touch and key interface, which looks the same on the L2 Pro, the Heltec kit, the ThinkNode M9 and the T-Deck. Where the way you do something differs, the manual shows it side by side:

TOUCH

TAP, HOLD, SWIPE on the L2 Pro, the Heltec kit and the T-Deck's screen.

KEYS · M9 / T-DECK

A focus ring marks the selected item. Keys are shown as `OK` `BACK` `←` `→` `↑` `↓` `MSG` `@`.

OLED BOARDS

The joystick on the Wio L1 family, or the single button on the T096. See chapter 17.

Names in **bold white** are exactly what the screen shows. A path such as **Settings › Radio › Scope** means: open the Settings tab, tap the Radio card, then the Scope card. Screens in this manual were captured on real devices; each caption says which board and, where it matters, which build.

02 / NEW IN 118 NEW

What Cairn 118 changes.

Cairn 118 moves Cairn onto MeshCore 1.18.0, fixes regions for messages you type on the device, adds delivery, resend, automatic retry, copy and paste to chats, lets alerts play your own sounds from the card, and gives the M9 and T-Deck a text box instead of an on-screen keyboard.

CORE · EVERY BOARD

MeshCore 1.18.0 underneath

Cairn now runs on upstream MeshCore 1.18.0, with its fixes: a guard against malformed packets, scope code 0 read as no scope, and properly spaced ACKs on direct routes through repeaters.

CHATS · SCOPE

Regions for device messages

Settings › Radio › **Scope** sets the default region, and each channel can have its own. Regional repeaters no longer drop what you type on the device. More

CHATS

Delivery and resend

A direct message shows **sending…**, then two ticks, or **not delivered** with a red **!**. **Resend** sends it again as the same message. More

CHATS

Auto retry

Off by default. Up to 3 sends by the path, then up to 2 floods, spaced over about two and a half minutes. More

CHATS

Copy and paste

Hold a message, tap **Copy**; any text field then offers **Paste**. More

CHATS

Path in the chat

A direct chat's ⋯ page shows the path, with **Reset path** and **Find path**. More

AUDIO · L2 PRO, T-DECK

Your own alert sounds

MP3 or WAV files in `/Sounds` on the card can replace the message, channel, advert, sent and startup tones. More

M9 · T-DECK

A text box, not a keyboard

The device's own keys type into a tall text box. A slim strip holds Emoji, **?123**, the bytes left and **Send**. More

M9 · T-DECK

⋯ is one key away

→ from the message field jumps to the chat's ⋯. The ring now outlines a message's bubble. More

T-DECK PLUS

GPS from boot

The GPS no longer stays silent after a restart, Quectel L76K units keep their GPS, and the clock sets itself from GPS again.

HELTEC V4 TOUCH

Battery voltage

The battery reading no longer runs about 1 % high.

Tested so far

- **T-Deck Plus (u-blox GPS):** the GPS fix after boot and after GPS off/on, the new text box, typing, and a direct-message alert playing a chosen MP3 from the card.
- **ThinkNode M9:** the screens in this chapter's chat sections were captured on an M9 running 117 beta: delivery ticks, auto retry and not delivered.
- **Built and unit-tested, not yet tried on hardware:** a scoped message passing a regional repeater, paste, an L76K T-Deck, alert sounds on the L2 Pro, and the SF32 and Wio L1 sends.

03 / INSTALL AND UPDATE

Getting Cairn onto the board.

Each board family installs a different way. Updates keep your identity, contacts, channels and settings. A first install on an ESP32 board erases everything.

YOU WANT TO…	ESP32 BOARDS	NRF52840 BOARDS	T-DISPLAY SF32
First install	`-full.bin` at `0x0`, after a full erase	the board's `.uf2`	the `-install.zip`: unzip, double-click
Update	online, Wi-Fi transfer, or `-update.bin` at `0x10000`	the board's `.uf2` again	online, or the `-install.zip` again
Recover	`-full.bin` at `0x0`, after a full erase	copy the `.uf2` again	hold A for 12 s, run the installer again
Boards	L2 Pro, Heltec V4 kit, ThinkNode M9, T-Deck, Heltec repeaters	Wio L1 Pro, DV1, L1 e-ink, T096, T114 repeater	T-Display SF32

ERASES

A first install on an ESP32 board erases the whole flash: the node's private key, contacts, channels and settings. If the board already runs MeshCore and you want to keep its identity, export the private key and contacts in the MeshCore app first.

ESP32 boards: web flasher or esptool

Write the `-full.bin` *or* the `-update.bin`, never both. Coming from stock MeshCore, Meshtastic or the maker's firmware, use the `-full.bin`.

01 Open Espressif's web flasher in Chrome or Edge (espressif.github.io/esptool-js) and click **Connect**. Pick the board's serial port.

02 **First install:** click **Erase Flash** and wait. Set the address to `0x0`, choose the `-full.bin` and click **Program**.
Update: set the address to `0x10000`, choose the `-update.bin` and click **Program**. Don't erase.

03 Click **Disconnect**, then tap Reset on the board.

The same with esptool v5 (`pip install esptool`):

```
# first install or recovery
esptool --chip esp32s3 erase-flash
esptool --chip esp32s3 -b 921600 write-flash 0x0 MeshCore-<Board>-CairnN-<date>-full.bin

# update
esptool --chip esp32s3 -b 921600 write-flash 0x10000 MeshCore-<Board>-CairnN-<date>-update.bin
```

Download mode

BOARD	HOW
Wio Tracker L2 Pro	hold `User/Boot`, tap `RST`, release `User/Boot`
Heltec V4 kit	hold `PRG`, tap `RST`, release `PRG`. Don't hold IO at the same time: IO is a strap pin.
ThinkNode M9	automatic through its USB-serial chip; it has no Boot button. If it won't connect, replug the cable. On Windows, install WCH's CH340 driver if no COM port appears.
T-Deck / T-Deck Plus	usually automatic. If not: switch it off, hold the trackball in (it is BOOT), switch it on, release.

SECOND SLOT

A board that installed its last update over Wi-Fi may be running from its second app slot, and a USB write at `0x10000` would then be ignored at boot. Clear the boot selector first; this keeps your data:

```
esptool --chip esp32s3 erase-region 0xe000 0x2000
```

nRF52840 boards: copy a UF2

01 Connect the board with a data cable (charge-only cables are common).

02 Double-tap `Reset` (`RST` on the T096). A USB drive appears.

03 Copy the board's `.uf2` onto the drive. The board reboots by itself when the copy finishes, and the drive disappears.

04 Check the splash: it shows **BUILD N** for the release you copied.

A good copy takes several seconds. If Finder stops with error **-36** on a Mac, copy from Terminal instead: `cp -X MeshCore-<Board>-CairnN-<date>.uf2 /Volumes/<drive>/`. Copying never touches the bootloader, so a bad copy is fixed by copying again.

T-Display SF32: the installer

01 Slide the power switch **ON** with the battery connected. With it off and only USB, the board restarts every 20 to 40 seconds once the flash tool takes over.

02 Take the core off the keyboard and plug it into the computer with a USB-C data cable.

03 Unzip `MeshCore-TDisplaySF32-CairnN-<date>-install.zip` and run `Install on Mac.command`, `Install on Windows.bat` or `./install-linux.sh`.

04 Leave the board alone until the installer says **Done**.

On a Mac the installer needs Python 3 (`xcode-select --install`). The SF32's download mode is in the chip's ROM, so a failed install can always be run again.

> TIP
> After any update, glance at the splash: **BUILD N** confirms the build you installed. **Settings › System › About** shows it too, with the slot (**app0 · USB**, **app1 · OTA**).

Update over Wi-Fi from the board

On the L2 Pro, the Heltec kit, the M9, the T-Deck and the SF32 (docked on its keyboard, which carries the Wi-Fi chip):

01 Join Wi-Fi: **Settings › Connect › Wi-Fi**.

02 Open **Settings › System › Firmware** (the page is titled **Firmware update**) and tap **Updates**

to check.

03 When it reads **Cairn N available**, open it and tap **Install**.

The board downloads its own `-update.bin` over HTTPS and checks its size, SHA-256 and board marker before installing. The new build starts on trial and goes back to the old one by itself if it fails to start. While on Wi-Fi the board also checks once a day and shows a notice when a newer build is out. Nothing installs without your confirmation.

BETAS

Online updates read only the latest full release; pre-releases (betas) are never offered. Install a beta by hand: Wi-Fi transfer, USB, or UF2.

Wi-Fi transfer (from a browser)

Called Updater mode before Cairn 116. Use it to send a firmware file from a computer on the same network, and on the L2 Pro and T-Deck to add music or alert sounds to the card.

01 Join Wi-Fi, then tap **Wi-Fi transfer** on the Firmware update page. On the L2 Pro and T-Deck you can also start it from **Settings › Hardware › Storage**.

02 The board shows an address and a password. Open the address in a browser on your network and type the password.

03 Upload the board's `-update.bin`, or drop MP3 and WAV files to save them in `Music` (optionally in an album folder) or `Sounds`.

Bluetooth is off while Wi-Fi transfer runs, so a phone linked over Bluetooth disconnects. After a firmware upload the board restarts into the new build. A new password is made for every session.

04 / FIRST START

Ten minutes to on the air.

Name the node, match your local radio settings, join your channels and announce yourself. Everything here can be changed later.

The splash

For about three seconds at power-on the splash shows the Cairn mark, **BUILD N**, when the build was made, and three columns: the MeshCore version, the board and the battery. A thin bar along the bottom counts down to Home.

The first boot after a full erase sits on **Loading…** for a while as it formats storage and creates the node's identity. Leave it alone.

SPLASH

Don't hold the User button while the splash is up. A long press there starts the USB rescue console (**Rescue CLI (USB) · restart to leave**); restart to leave it.



T-DECK The splash names the build. Quote it in bug reports.

First-run checklist

01 **Name your node.** **Settings › Radio › Profile › Node name**, up to 31 bytes, emoji allowed. Turn **Position** on to share your location in adverts. Save asks **Send advert now?**; choose **Not now** until the radio is set.

02 **Match the radio.** **Settings › Radio › Radio**: pick the **Preset** your local mesh uses (**USA/CAN**, **EU/UK 869**, **EU classic**) or set a custom frequency, bandwidth, spreading factor and coding rate. Keep **TX power** within your region's limit. Save asks **Apply radio change?**.

03 **Set a scope if your mesh uses regions.** **Settings › Radio › Scope.** Ask a local operator; many meshes need none. See Scope.

04 **Join your channels.** **Settings › Radio › Channels › Join.** Public is the channel every MeshCore node starts with.

05 **Pair the phone app** if you use one: open the MeshCore app, connect over Bluetooth and type the PIN shown on Home's Network card or in **Settings › Connect › Bluetooth**.

06 **Choose a GPS mode.** **Settings › Location › GPS**: **Eco** for days of battery, **On** for tracking.

07 **Announce yourself.** On the Radios tab tap **Advert**, choose **Zero-hop** or **Flood** and tap **Send**.

CLOCK

After a power cut or a fresh install the clock reads **--:--** until GPS, the phone app or Wi-Fi sets it. Logging in to a repeater needs a set clock.

TIP

The Bluetooth PIN is new at every boot unless you set a fixed one under **Settings › Connect › Bluetooth › Change PIN**. Six zeros switch back to an automatic PIN.

05 / THE SCREEN

Rail, bar, page.

Most pages have three parts: the icon rail down the left, the status bar along the top, and the content. Text entry pages hide the rail; the splash and the lock screen fill the screen.

The rail

Six tabs, top to bottom. The open tab gets a tinted pill in its colour, and every page title is drawn in its tab's colour.

TAB	COLOUR	MARKER
Home	teal	none
Chats	blue	red pill: unread messages, **1** to **99+**
Nodes	violet	magenta pill: new adverts since you last looked, **+1** to **+99**
Radios	amber	amber up-arrow when a packet is sent; teal dot when one is received; amber dot until the noise floor is calibrated; red **PAUSED** while a band scan holds the radio
GPS	green	satellite count (green with a fix, dim while searching); a moon while Eco sleeps; the icon turns red while GPS is off
Settings	grey	amber dot while Wi-Fi transfer runs; teal dot while the card is busy with a background delete

TOUCH

- TAP a tab to open it at its root.
- TAP the open tab on a sub-page: back to its root, your place kept. Tap again: scroll to the top.
- HOLD Chats while unread: **Mark all read?**

KEYS

- ← from the left edge of a page reaches the rail; ↑ ↓ move along it, OK opens.
- → off the rail lands on the page.

- M9: HOME and MSG jump straight to Home and Chats.

OLED BOARDS

Page tabs run along the bottom, or down the side if you set **Settings › Display › Page tabs** to Side. See chapter 17.

The status bar

The page title sits on the left, in its tab's colour. The clock is always centred; it reads **--:--** until the time is set. On the right, from the outside in: the battery (percent inside the outline, a bolt while charging, a plug on USB), then Wi-Fi bars, Bluetooth, mute and the padlock. The Bluetooth icon turns teal when the app is linked over Bluetooth; Wi-Fi turns teal when the app is linked over Wi-Fi and amber during Wi-Fi transfer. On the L2 Pro and T-Deck a **MIC** or **REC** pill shows whenever the microphone is live.

The bar's bottom edge flashes amber for a moment when the radio sends and teal when it receives.

Notices. Short messages take over the status bar for a moment, led by a coloured mark (red for errors), then the bar comes back. A notice with a chevron, such as **New message from Bob**, opens what it is about when you tap it.

TIP

Tap the clock, or anything to its right, to open the quick panel from any page. On a tab's root page the title opens it too.

Going back

A sub-page shows a teal ‹ before its title. Any of these goes back one page; unsaved edits ask **Discard changes?** first.

TOUCH

- TAP the ‹ **Title** zone, or just below it.
- SWIPE right, mostly sideways, starting well in from the left edge (not on the rail). Works even if the swipe starts on a button.
- The User / PRG button's long press (the default **Back**).

KEYS

- M9: BACK. It first leaves adjust or pan mode, closes a menu, then goes back. At a tab's root it goes Home.
- T-Deck: ⌫ outside a text field, or hold the trackball in with nothing selected.

OLED BOARDS

Wio L1: the Back button. T096: pages loop forward with a click.

The back swipe does nothing on root pages, on the map (dragging pans it), on sliders, and on the keyboard screen.

Gestures

GESTURE	HOW	DOES
Tap	Touch and lift within about 0.6 s without moving.	Selects or activates. A highlight shows while you press.
Hold	Keep still for about half a second.	The control's hold action, while your finger is still down. A short tick sounds when a hold opens a menu or page. A control with no hold action takes a hold as a tap.
Drag, flick	Move your finger.	Lists follow your finger; a flick keeps them moving.
Back swipe	Swipe right at least 60 px, mostly sideways, starting away from the rail.	Back one page.
Pull down	At the top of the Nodes list, pull down until it says **Release to scan**.	Repeater scan.
Swipe up	On the quick panel.	Closes it.
Pinch	Two fingers, on the GPS map only.	Zooms. Elsewhere a second finger is ignored, so a resting thumb won't cancel a tap.
Double tap	On the GPS map.	Zooms in one level on that point.

There is no sideways swipe between tabs.

Dialogs, card pages and holds

- **Dialogs** put the safe choice on the left, and it is the default. Tapping outside, pressing the User button or locking chooses it. Taps are ignored for a moment after a dialog opens, and its confirm button never appears under the finger that asked.
- **Menus are card pages.** Holding a message, a chat, a node or the map opens a page of cards in the Settings style, each with a title and a live summary. A red card (such as **Delete…**) sits last.

- **Hold to confirm.** Tiles that erase something say how long to hold (**Hold 1.5 s**, **Hold 2 s**) and fill as you hold; a dialog then asks once more. Sliding your finger cancels.
- **Asks before switching.** Turning Wi-Fi, Bluetooth or GPS on or off from the board asks first, with a line that says what changes (for example that a linked app will disconnect). Switching GPS between On and Eco acts at once.

The quick panel

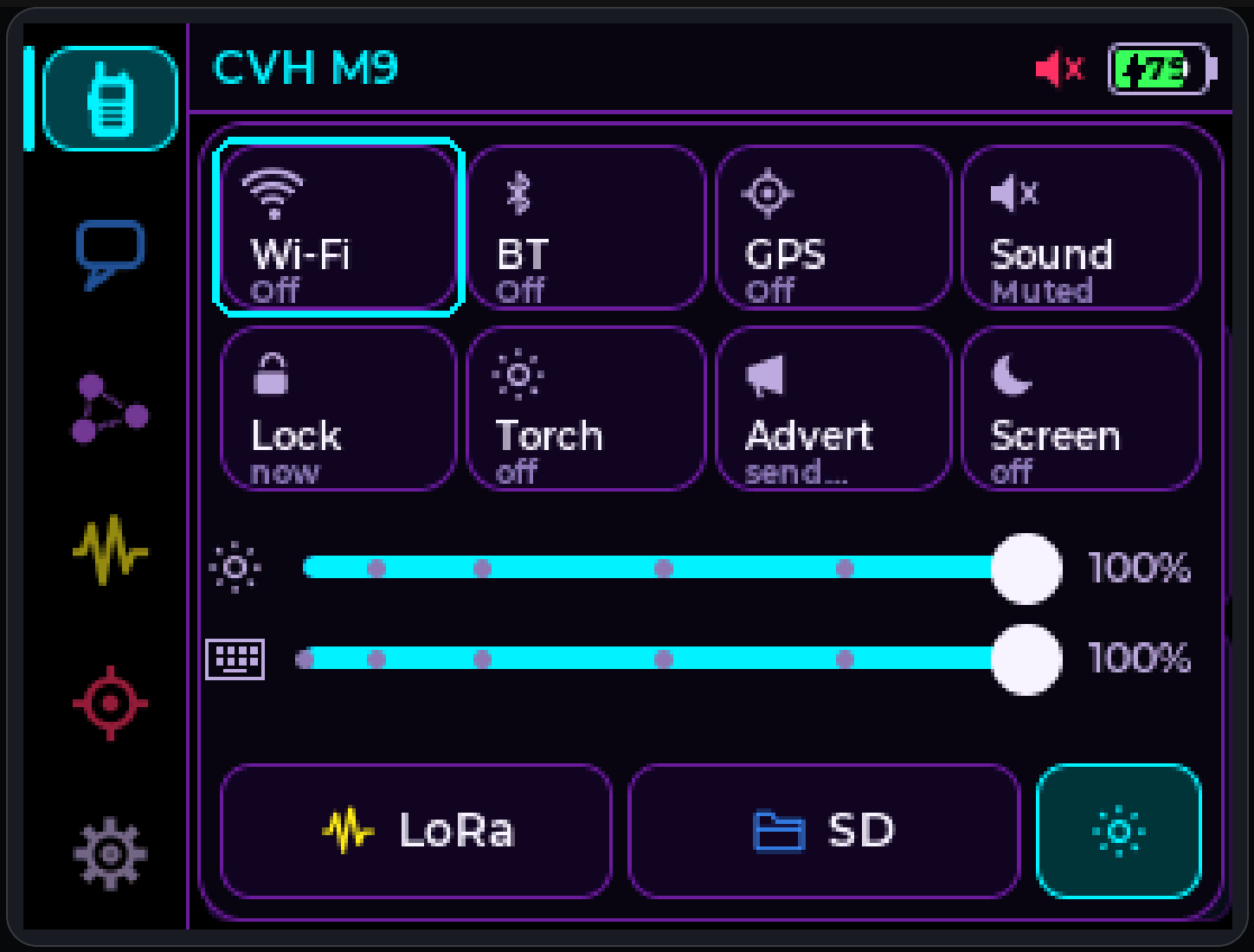
One place for the switches you reach for most.

OPEN

- Tap the status bar's clock or icons.
- Press the User / PRG button once (the default click).
- L2 Pro, Heltec kit, T-Deck: tap Home's big clock.
- M9: `CTRL`. T-Deck: `Alt` + `C`, or `Space` (the default click action).

CLOSE

- Swipe up on it, or tap outside it.
- Press the User button again.
- M9: `CTRL` again, or `BACK`.
- Locking, or a message taking over, also closes it.



M9 Ring on the Wi-Fi tile. The second slider is the key light.

TILE	TAP	HOLD
Wi-Fi	On or off, after **Turn Wi-Fi on?** / **off?**	Wi-Fi settings
BT	On or off, after a confirm	Bluetooth settings
GPS	On → Eco → Off → On. Leaving Eco for Off, and Off for On, ask first.	GPS settings
Sound	Mute or unmute	Sound settings
Lock	Lock now	same as tap
Torch	The whole screen white at full brightness. A tap or a press ends it.	same as tap
Advert	The Zero-hop / Flood advert sheet. Nothing is sent until **Send**.	same as tap
Screen	Screen off (and lock, if Auto lock is on)	same as tap

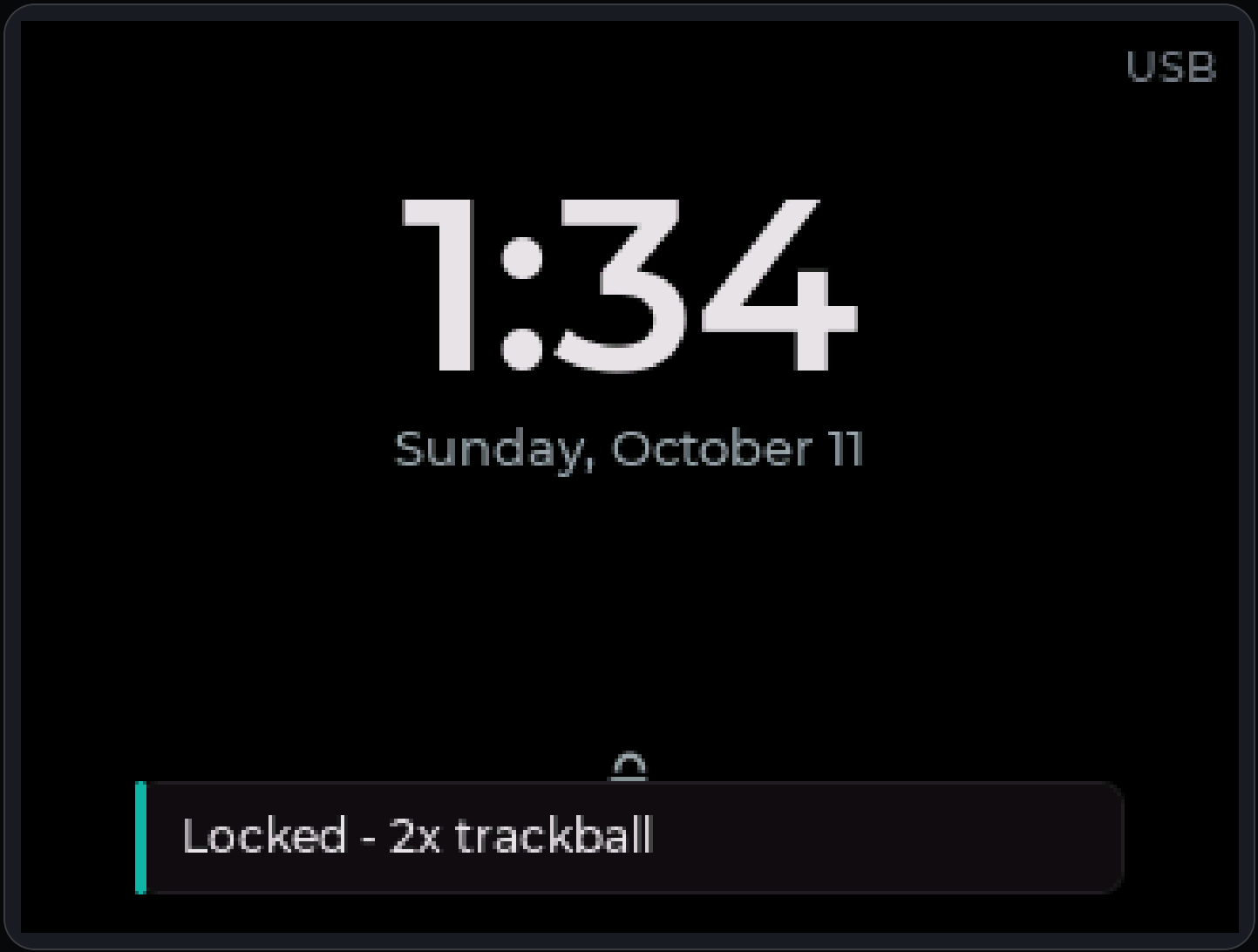
Below the tiles: a **brightness** slider, a **key light** slider on the M9 and T-Deck, and a **volume** slider on boards with a speaker (it reads **Off** while muted). The bottom row holds **Audio** (L2 Pro and T-Deck), **LoRa** (the radio settings), **SD** (Files) and **Capture**, which saves a screenshot of the page under the panel to `/Screenshots` on the card.

> TIP
> Hold a toggle tile to jump to its settings page. **‹** then returns you to the page you were on, not to Settings.

Lock and wake

Locking keeps touches and presses from doing anything by accident. The lock screen shows a large clock, the date, the battery, a card for unread messages, and a padlock with the unlock hint.

BOARD	LOCK	UNLOCK
L2 Pro	hold `WAKE` 1 s, the quick panel's Lock, or Auto lock	double-press `WAKE` or `User`
Heltec kit	hold `IO` 1 s, Lock, or Auto lock	double-press `IO` or `PRG`
ThinkNode M9	the quick panel's Lock (`CTRL`), or Auto lock	`OK` twice, or hold `OK`
T-Deck	the quick panel's Lock, or Auto lock	click the trackball twice



`T-DECK · 113` The lock notice names the unlock key.

With **Touch unlock** on (**Settings › Hardware › Lock**), holding the padlock for 1.5 s also unlocks; a ring fills around it. While locked, other presses only light the lock screen.

Waking. Any button or key wakes a dark screen and does nothing else. With **Tap to wake** on, a touch wakes it too. On the L2 Pro and Heltec kit a `WAKE` / `IO` click turns a lit screen off. On the M9, `MIC` turns the screen off.

> TIP
> A double-press of WAKE (or User) unlocks straight from a dark screen. With Auto lock on, that is the one gesture you need to pull the device out of a pocket.

Screensaver and themes

Screensaver (**Settings › Appearance › Screensaver**) replaces a screen that would otherwise stay lit, that is, with **Timeout** set to Never or with **Stay on USB**. Styles: **Beacon** (a small card with the time and new messages that moves every 30 s), **Ambient** (clock, date, sun times and status at a glance), **Night clock** (a large clock, red after sunset), **Auto** and **Off**. It starts after 1 to 30 minutes without a touch, runs on USB only unless you allow battery, and can show a message count, the sender or the full text. Any touch or key only ends it. New devices start with Ambient after 1 minute at 50 %.

Themes (**Settings › Appearance › Theme**): Dark, Neon, Light, Sunlight (highest contrast, for outdoors), Night (reds and ambers, keeps night vision), Ocean, Cartoon, Red Radar and Green Radar. Each previews live as you tap it.

06 / SHORTCUTS AND GESTURES

The cheat sheet.

The same action, three ways. Touch covers the L2 Pro, the Heltec kit and the T-Deck's screen. Keys cover the ThinkNode M9 and the T-Deck's keyboard and trackball. OLED covers the Wio L1 family's joystick and the T096's one button.

`OK` M9 d-pad centre · T-Deck trackball click

`Hold OK` long press of what is selected

`User` L2 Pro · `PRG` Heltec

`WAKE` L2 Pro · `IO` Heltec

Moving around

ACTION	TOUCH	KEYS (M9 / T-DECK)	OLED BOARDS
Back one page	Tap **‹ Title**; SWIPE right from away from the rail; User long press	M9 BACK. T-Deck ⌫ outside a text field, or hold the trackball with nothing selected	L1: Back button
Home	Tap the Home tab	M9 HOME; BACK at a tab's root	L1: ← → to the Clock page
Next / previous page	Tap a rail tab	← onto the rail, ↑ ↓, OK	L1: ← → (bottom tabs) or ↑ ↓ (side tabs). T096: click
Back to a tab's root, then the top	Tap the open tab, then again	Rail cell, OK	—
Scroll a long list fast	FLICK	Quick repeated ↓: after six presses each moves 2 rows, then 4. T-Deck: roll the ball fast	—
Open the quick panel	Tap the status bar clock or icons; User click; L2 / Heltec / T-Deck: tap Home's big clock	M9 CTRL. T-Deck Alt + C or Space	—
Close the quick panel	SWIPE up, tap outside, or User	CTRL again, or BACK	—
A quick tile's settings	HOLD the tile	Ring on the tile, Hold OK	—
Long press anything	HOLD ½ s	Hold OK	T096: hold the button
Change a slider or stepper	Drag; hold − / + to repeat	← → step it; OK enters adjust mode (amber ring), then ↑ is more	L1: ← → in the editor

Messages

ACTION	TOUCH	KEYS (M9 / T-DECK)	OLED BOARDS
Open Chats	Chats tab	M9 `MSG`	L1: Messages page
New message	`User` double-click; **+ New message**	M9 `@`. T-Deck double `Space`	L1: Messages, pick a recipient, then a preset
Find a recipient	Tap **Search...**	Just type in the picker	—
Start a reply in a chat	Tap the message field	Type a letter: the text box opens with it	L1: reply with a preset
Send	**Send** beside the field	M9 `OK` (nothing selected). T-Deck keyboard `Enter`	—
Message menu	`HOLD` the bubble	Ring on the bubble, `Hold OK`	—
Chat menu ⋯	Tap ⋯ at the top right	`→` from the message field or Send	—
Read a long message	Tap the bubble (the Reader)	Ring on it, `OK`	L1: open the message
Mark everything read	`HOLD` the Chats tab	Ring on the Chats tab, `Hold OK`	—
Emoji	☺ key; **Recent** keeps your last 24	☺ in the strip; press again to close	—
Caps lock	Double-tap Shift	The keyboard's own Shift	—
Fix a near-miss key	Slide to the right key before lifting	—	—
Delete fast	Hold `⌫` to repeat	M9: hold `DEL` deletes the word before the cursor	—

ACTION	TOUCH	KEYS (M9 / T-DECK)	OLED BOARDS
Leave a field, keep the draft	‹ or `User` long press	M9 `BACK`. T-Deck: hold the trackball, or `⌫` on an empty field	—

Mesh, radio and map

ACTION	TOUCH	KEYS (M9 / T-DECK)	OLED BOARDS
Send an advert	HOLD Home's LoRa card; quick panel **Advert**; Radios › **Advert**	M9 ADV. Radios tab: the ring lands on Advert, so OK OK sends zero-hop	T096 and repeaters: hold on Spectrum
Scan for repeaters	PULL DOWN at the top of Nodes	↑ past the top of Nodes	—
Node menu	HOLD a node	Hold OK	—
GPS on / off	Quick panel GPS tile; the GPS page's switch	M9: hold ADV	L1: Settings › Location
Open the map	GPS tab › **Map**	M9 MAP	—
Pan the map	DRAG	M9 MAP again: pan mode, arrows pan, OK taps the centre	—
Zoom the map	PINCH, double tap, or + / −	+ / - keys	—
Recentre on you	Tap the GPS tab again, or Follow	Follow button	—
Map menu	HOLD an empty part of the map	—	—
Pick a Radios tab	Tap LoRa / Wi-Fi / Bluetooth	T-Deck 1 2 3	—
Scanner Plot / List	Tap the switch	#	—
Clear band-scan peaks	HOLD the band plot	—	—

ACTION	TOUCH	KEYS (M9 / T-DECK)	OLED BOARDS
Read a moment on the Battery graph	Tap the graph	`OK` on the graph: a cursor at now; `←` `→` move it, `BACK` clears it	—
GPS views	Tap the status or any card for GPS data	Ring on a card, `OK`	L1: joystick click steps through position, compass and GPS data

Screen, sound and power

ACTION	TOUCH	KEYS (M9 / T-DECK)	OLED BOARDS
Wake the screen	Any button; a tap with Tap to wake on	Any key (it only wakes)	Any press (it only wakes)
Screen off	`WAKE` / `IO` click; quick panel **Screen**	M9 `MIC`; quick panel	—
Lock now	Hold `WAKE` / `IO` 1 s; quick panel **Lock**	Quick panel **Lock**	—
Unlock	Double-press `WAKE` or `User`; hold the padlock 1.5 s (Touch unlock)	M9 `OK` twice or `Hold OK`. T-Deck: trackball twice	—
Mute / unmute	`User` triple-click; quick panel **Sound**	Quick panel **Sound**	—
Flashlight	Quick panel **Torch**; a button action	Quick panel **Torch**	—
Screenshot	Quick panel **Capture**; the **Screenshot** button action	Quick panel **Capture**	—
Clock and time zone	`HOLD` Home's header	Settings › System › Clock	L1: Settings › System
Power back on after Hibernate	`User` / `PRG` (not WAKE / IO)	M9: the power slider (off, then on) or RESET. T-Deck: the trackball click is not yet confirmed	—
Rescue console	Long-press `User` during the splash	—	T096: hold during the first 8 s
Player (T-Deck keyboard)	Tap the controls	`Space` play/pause, `n` / `p` next/previous, `s` shuffle, `r` repeat, `←` `→` seek 10 s, `↑` `↓` volume, `0` – `9` jump	—

ACTION	TOUCH	KEYS (M9 / T-DECK)	OLED BOARDS
Recorder (T-Deck)	Tap Record	`Mic` opens it; `r` or `Enter` record/stop, `m` mark	—

The configurable button

On the L2 Pro (`User`) and Heltec kit (`PRG`) the button's click, double-click and long press run the actions you pick in **Settings › Hardware › Buttons**. On the T-Deck the same page sets `Space` and double `Space` (outside a text field).

Click
Default: Quick panel

Double-click
Default: New message

Long press (1 s)
Default: Back

Triple-click
Always mute / unmute

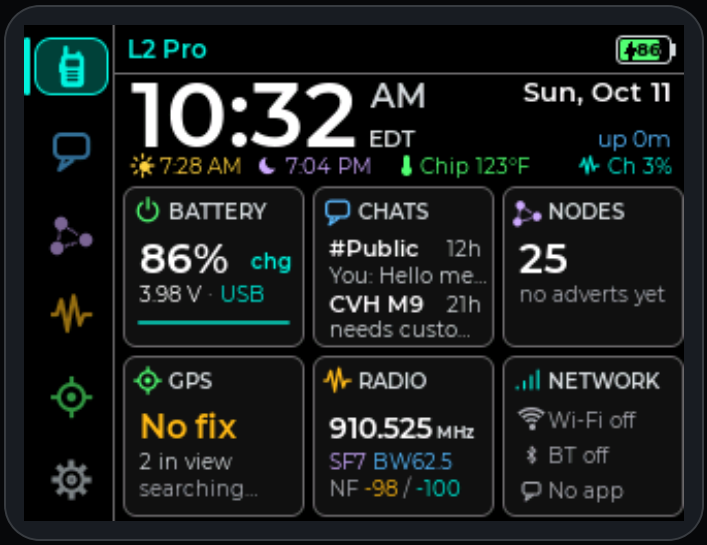
Actions to choose from, grouped General, Mesh, Tools and Audio: **None**, **Quick panel**, **Back**, **New message**, **Send advert**, **Ping last node**, **Scan repeaters**, **Toggle GPS**, **Flashlight**, **Screenshot**; on the Heltec kit also **Screen off** and **Lock now**; on the L2 Pro and T-Deck also **Player**, **Recorder** and **Play / pause**.

Fixed rules come first, in this order: while locked, a click or long press only lights the lock screen and a double-press unlocks; on a dark screen any press only wakes it; with the flashlight on, any press only turns it off; a triple-click always mutes; and while a panel, sheet or dialog is open, a press closes it instead of running its action.

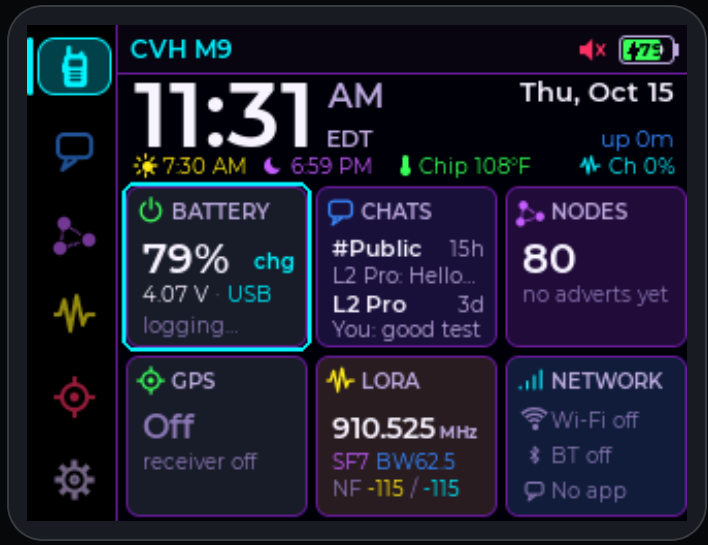
07 / HOME

Everything at a glance.

The first tab: a large clock over six live cards. Tap a card to open its page.



L2 PRO · 113 Home by touch. The node's name is the title.



M9 By keys, the ring starts on the Battery card.

Header and strip

The clock in your 12h, 24h or analog style, with the time zone; the date and the uptime on the right. Until the time is set it reads **--:--** and **time not synced**. The thin strip under it holds up to three readouts:

- **Sunrise and sunset** for today, from your GPS fix or your advert position. Hidden without a set clock and a position.
- **Temperature** from an environment sensor, or the chip's own sensor labelled **Chip**. The chip runs 30 to 40 °F above the room.
- **Channel busy**, **Ch 12%**: the share of the last minute the channel sat at least 6 dB above the noise floor. Amber above 30 %, red above 60 %.

> TIP
> Tap the big clock for the quick panel (L2 Pro, Heltec kit, T-Deck). Hold the header or strip for the Clock editor: style, time zone and daylight saving.

The cards

Battery

Percent, time left (**~14h**) or **chg** / **full** on USB, voltage and source, and a trend line.

TAP: BATTERY PAGE

Chats

Unread count and the two newest conversations with a one-line preview.

TAP: CHATS

Nodes

Saved contacts, **+N** new adverts, the last one heard with its age, distance and direction.

TAP: NODES

GPS

Fix with satellites and position, **No fix**, **Eco** with the next fix, or **Off**.

TAP: GPS

LoRa

Frequency, SF and bandwidth, noise floor and the current reading.

TAP: RADIOS · HOLD: SEND ADVERT

Network

Wi-Fi network, Bluetooth state or the pairing PIN, and how the app is linked (**App: BT**, **App: Wi-Fi**, **No app**).

TAP: SETTINGS › CONNECT

On the L2 Pro and T-Deck an **Audio** card sits one swipe down: songs, recordings and free recording time, or what is playing. While a song plays, Home's status bar shows its title.

TIP

The Network card is the quickest place to read the Bluetooth pairing PIN when you pair a new phone.

08 / CHATS

Read it, answer it, right there.

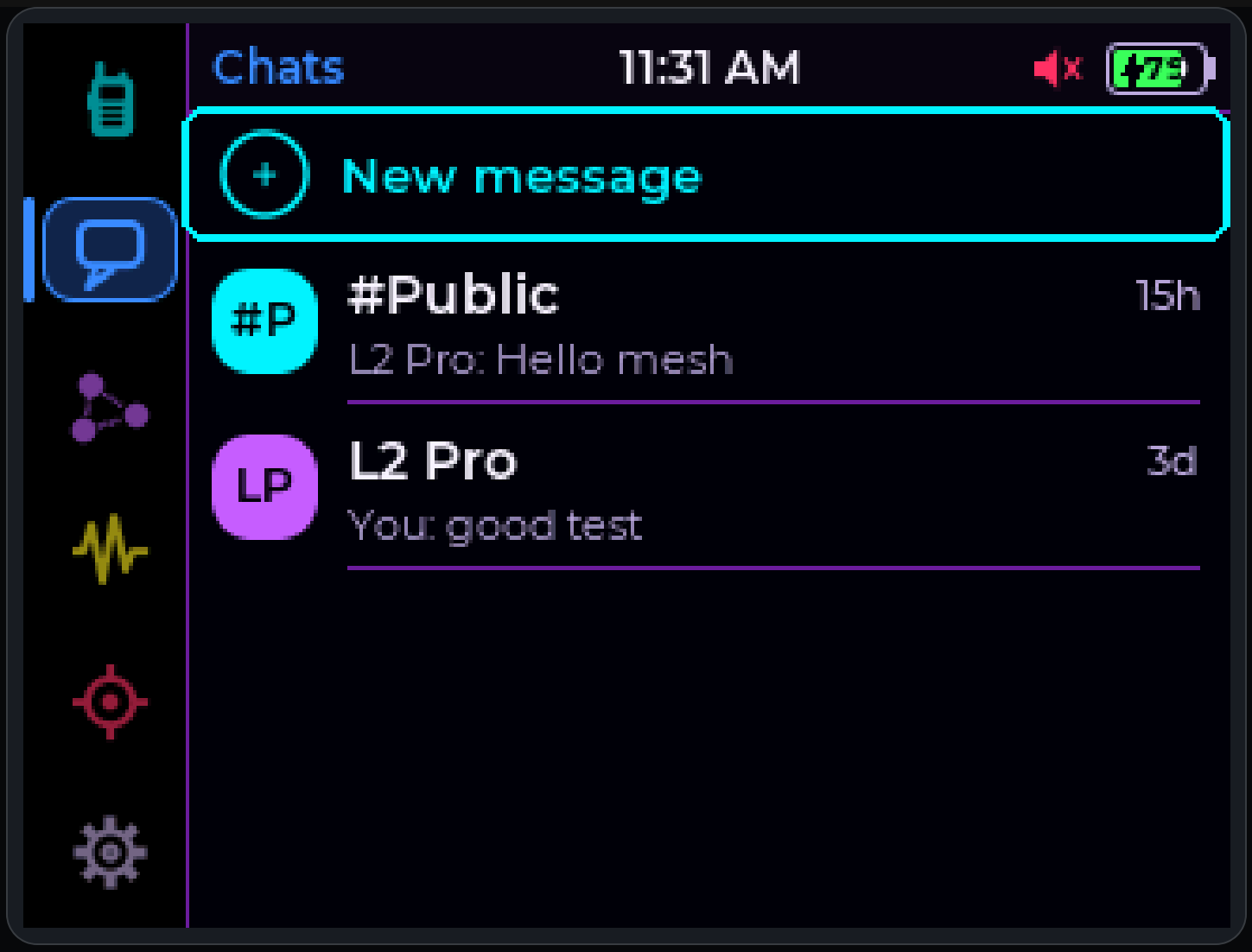
Direct messages, channels and room servers share one list. Threads show bubbles with hops, times, delivery and repeats heard. On the L2 Pro, Heltec kit, ThinkNode M9 and T-Deck.

The chat list

The title reads **Chats**, or **Chats · N unread**. **+ New message** sits at the top, then conversations, newest first. Each row has a coloured avatar, the name (in the accent colour while unread), the age of the newest message, a one-line preview (**You: …** for yours; channel posts start with the sender's name) and an unread pill.

- TAP a row to open the thread.
- HOLD a row for the chat's card page: **Open**, **Mark read**, **Contact** (route, favourite, share) or **Channel** (key and slot, leave), **All read…** and **Delete chat…**.

The list doesn't re-sort while your finger is on it. Channel names always start with **#**; rooms show as **@Name**.



M9 The list, ring on **New message**.

Starting a conversation

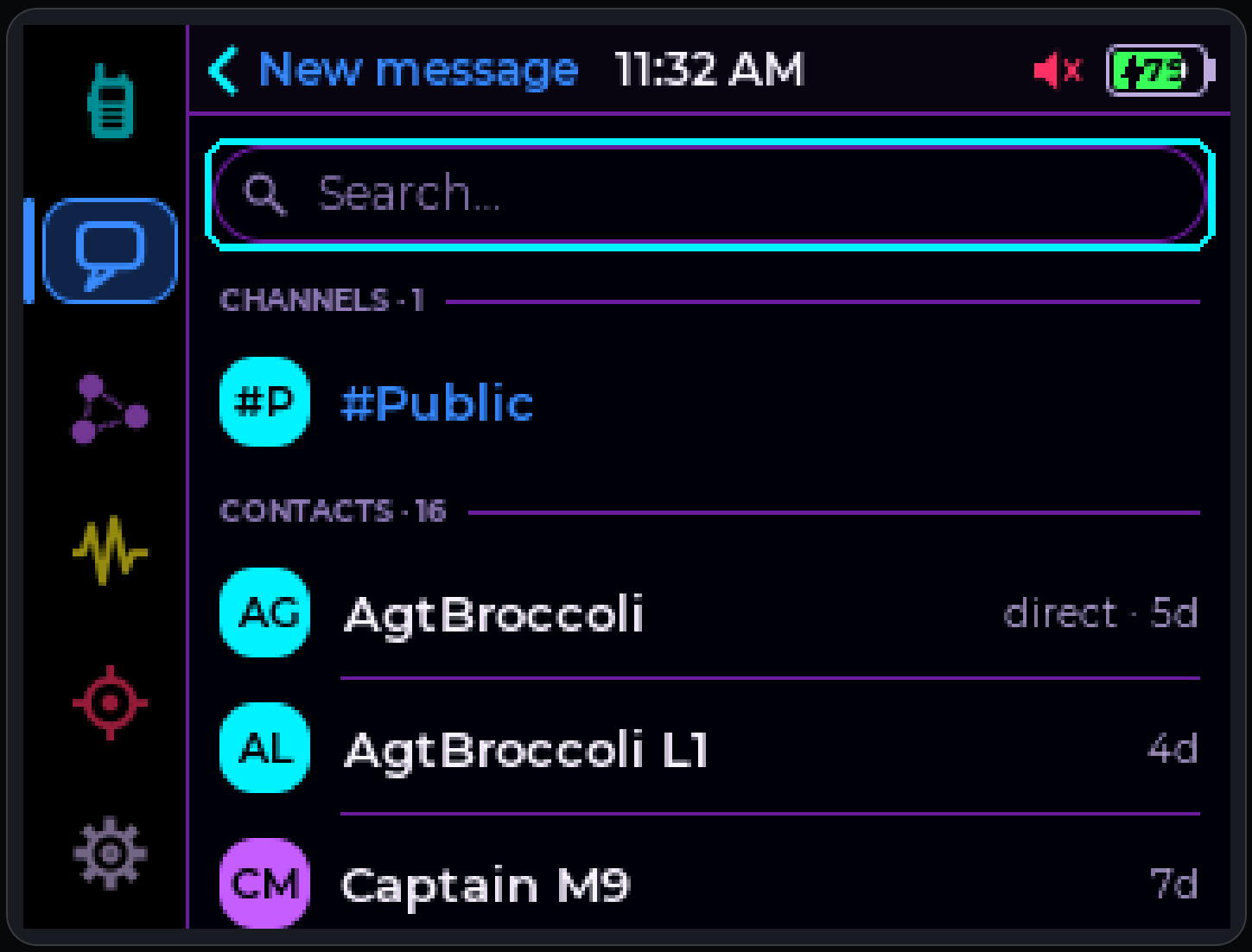
TOUCH

- **+ New message**, or double-click `User` / `PRG`.
- Nodes: hold a node › **Message**, or **Chat** on its detail page.
- Tap a **New message from…** notice.

KEYS

- M9 `@` opens the picker anywhere.
- T-Deck: double `Space` (default).
- In the picker, just type to search.

The picker lists **CHANNELS · N**, then **CONTACTS · N** with each contact's route and when it was last heard (**direct · 5d**). Rooms you are logged in to appear as **@Room**. Repeaters are never listed: use **Admin** on their node page. Pick a row and the thread opens ready to type.



M9 Opened with @.

Reading a thread

The title shows the name and message count (**#Public · 111 msgs**). Your bubbles sit on the right, others' on the left with the sender's name on channels. Date dividers (**Today**, **Yesterday**, **Sun Oct 11**) appear once the clock is set. Each bubble ends with a meta line: hops and time for incoming (**0 hops · 9:29 AM**), time and delivery for yours.

- TAP a bubble to read it whole in the **Reader**, with **‹ Prev**, **Actions** and **Next ›**. Each message you view is marked read.

- HOLD a bubble for the message's card page (below).
- Tap ⋯ at the top right for the chat's card page.

Mentions show as **@Name**; your own name is highlighted. Replies show the quoted line in grey with a bar in the margin.

Typing a message

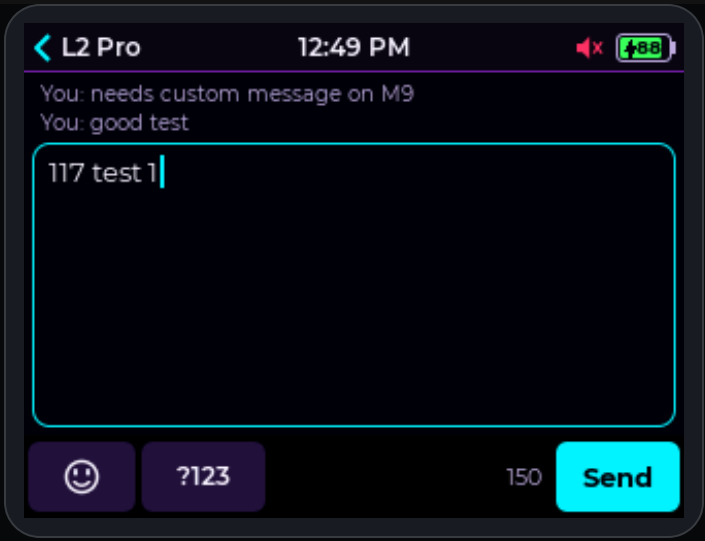
A direct message holds up to **160 bytes**. A channel post holds 160 bytes less your node name and two (it is sent as **Name: text**). Emoji and accented letters use more than one byte. A counter shows the bytes left once 20 or fewer remain, amber then red when over.

TOUCH KEYBOARD · L2 PRO, HELTEC

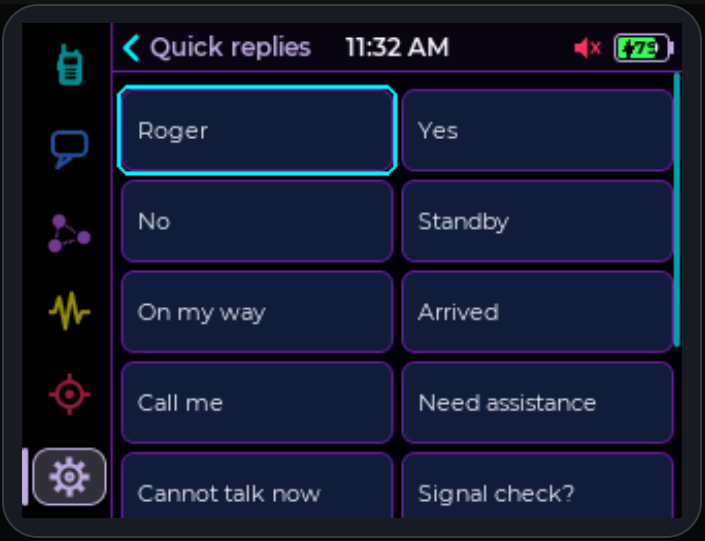
- Tap the message field: the full keyboard opens, with the last two messages above for context.
- A key enters when you lift; slide to fix a near miss first.
- **Send** sits beside the field, away from Delete. A slide only sends if it started on Send.
- Shift once for one capital; double-tap for caps lock.
- **?123** and **#+=** for symbols; ☺ for 229 emoji in seven tabs plus **Recent**.
- Hold Backspace to repeat; it removes a whole emoji at once.
- ‹ (or the User long press) leaves and keeps the draft.

TEXT BOX · M9, T-DECK 118

- No on-screen letters: your keyboard types into a tall box that wraps over most of the screen.
- In a thread, just start typing: the box opens with that letter.
- The strip below holds ☺ (emoji), **?123** (symbols the keys lack), the bytes left, and **Send** or **Done**.
- ☺ and ?123 open over the lower half; press again to close (T-Deck: or tap the text).
- Send: M9 OK with nothing selected; T-Deck Enter.
- M9: hold DEL to delete the word before the cursor.



M9 The text box: 150 bytes left.



M9 The ten quick replies, editable.

Quick replies. **Aa** beside the message field opens ten short presets as cards; a tap adds one to your draft. Edit them in **Settings › Radio › Quick replies**: each holds up to 32 bytes, and **Reset to defaults…** brings back Roger, Yes, No, Standby, On my way, Arrived, Call me, Need assistance, Cannot talk now and Signal check?.

Drafts. Each conversation keeps its own unsent draft. Drafts are lost at a restart.

Replying to a message

01 `HOLD` the message (keys: ring on the bubble, `Hold OK`) and choose **Reply**.

02 The keyboard opens with **Replying to Bob** and the quoted message. A bar above the message field shows the quote with an **×** to cancel reply mode and keep the draft.

03 Send. Replies use the phone apps' format, so they look the same there.

Quick reply on the same page picks a preset and asks **Send reply?** first.

Copy and paste 118

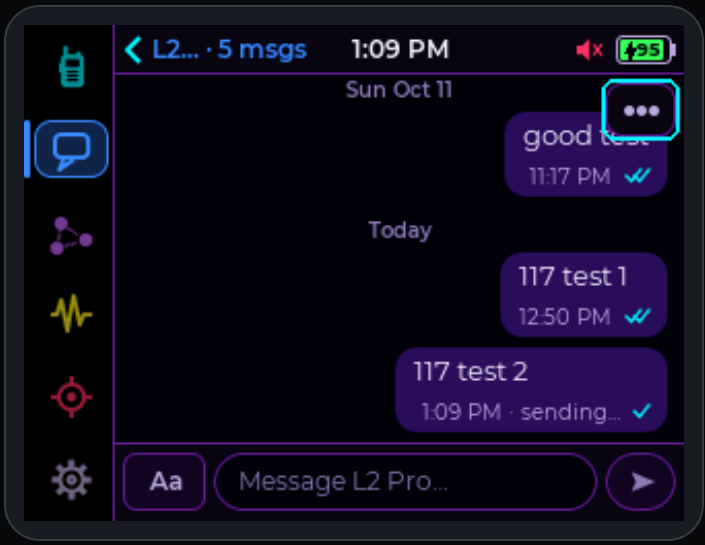
Hold a message › **Copy** (a channel message copies its text without the sender's name). Every text field then shows a **Paste** chip: beside the context lines on the full keyboard, in the strip on the M9 and T-Deck. Paste inserts what fits the field's limit; **Pasted · cut to fit** says when part was left out. The clipboard holds one entry until a restart.

TIP

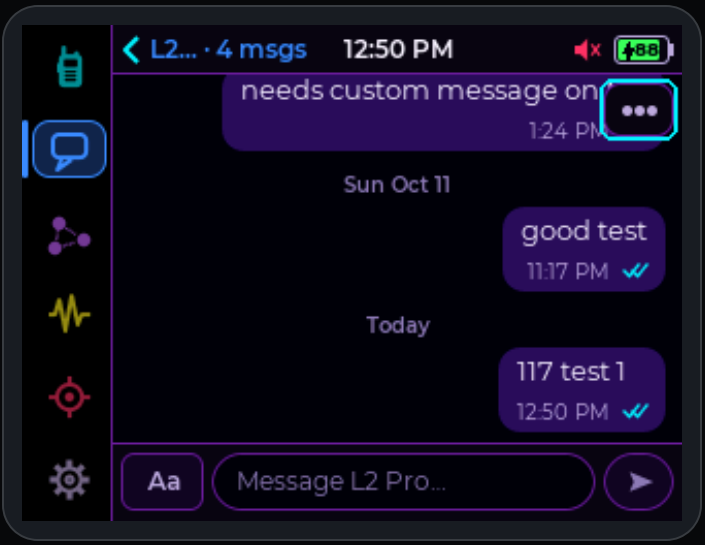
Copy works on anything in a thread, including a received channel key or a repeater name you want to type into another field. On the keys boards, reach the Paste chip with the arrows.

Delivery, heard and resend 118

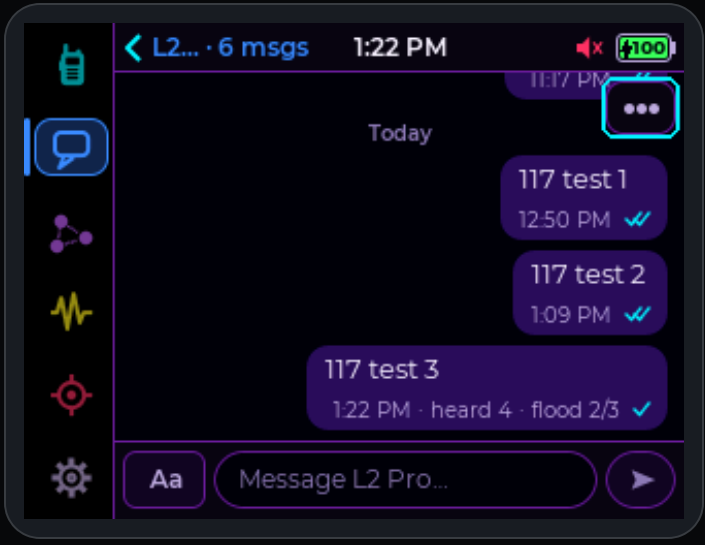
A direct message you send from the device shows how it is doing on its bubble:



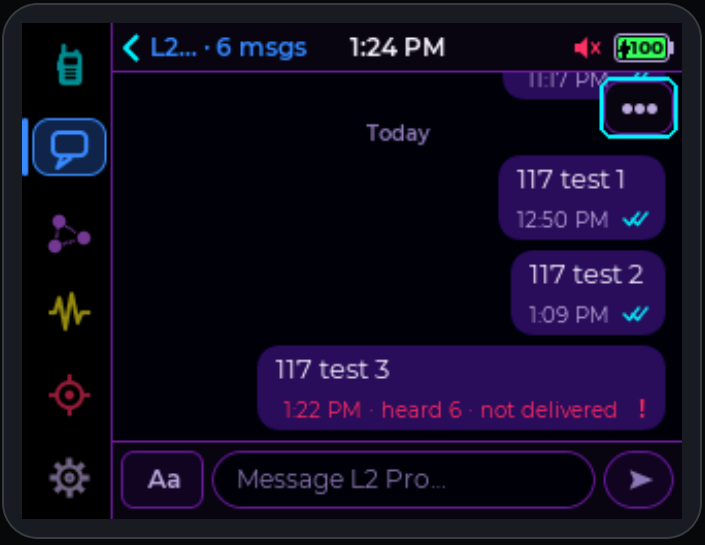
M9 **sending...** One tick: it left this device.



M9 Two ticks: the other side confirmed it.



M9 Auto retry at work: **heard 4 · flood 2/3**.



M9 **not delivered** and a red **!**.

- **Heard N.** Your messages count how many repeats of them this radio heard (**heard 3**): proof the mesh carried it, even on a channel, which has no delivery receipt.

- **Not delivered** means no confirmation came in time. It may still have arrived; a late confirmation still turns it to two ticks.
- **Resend**: hold your message › **Resend** (or in **Details**). A direct message goes again as the same message, so the other side shows it once. Your own channel message asks **Resend message?** and goes again as a new message with its own heard count.
- A Cairn device that receives the same direct message twice (a retry whose first copy arrived but whose confirmation was lost) shows it once.

Auto retry 118

Settings › Radio › Auto retry flips on with a tap (off by default; the card reads **3 path · 2 flood** while on). An unconfirmed direct message is then sent again by its path, 3 tries in all; then the path is reset and it floods up to 2 times. A contact with no known path gets 3 floods. Each wait is longer than the last (about 10 s, 20 s, 30 s, then a minute, each varied a little), so a run lasts about two and a half minutes without flooding the mesh. The bubble shows **retry 2/3** or **flood 1/2** while it works. The automatic tries make no sound.

AIRTIME

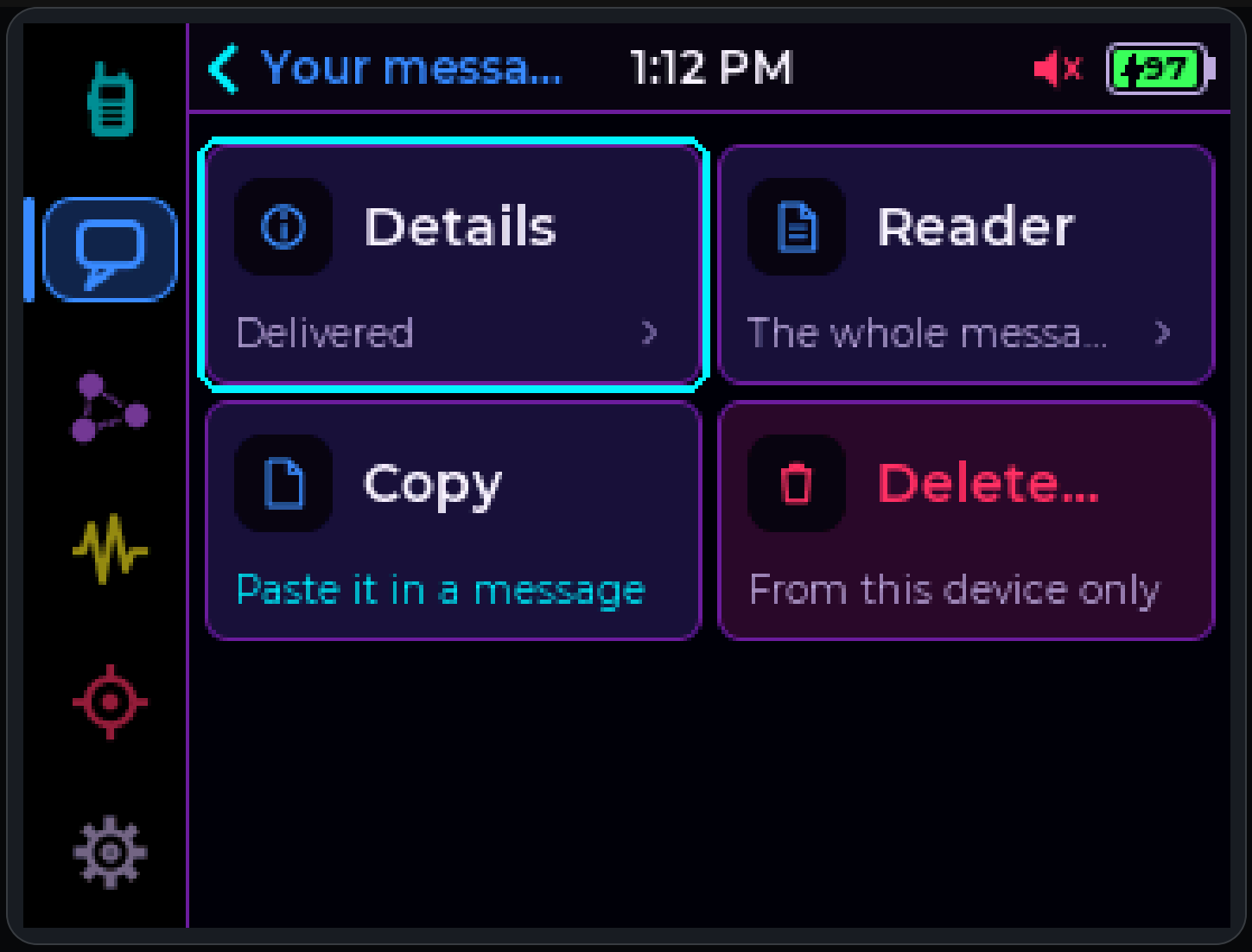
Floods reach every repeater in range. Leave Auto retry off on a busy or duty-limited mesh, and use Resend by hand when a message matters.

The message's card page

Hold a bubble to open it. Someone else's message offers **Reply**, **Quick reply**, **Details** (hops and signal), **Reader**, **Copy** and **Delete…**. Your own offers **Resend** (not on a delivered message), **Details**, **Reader**, **Copy** and **Delete…**.

Details is a page of cards. For an incoming message: who, where and when, then how it arrived: hops, signal (SNR and RSSI) and the route, naming repeaters where it can. For yours: to whom and whether it was delivered, then the repeats heard and up to four repeaters with their SNR.

Delete… asks first and removes the message from this device only; the phone app keeps its own copy.



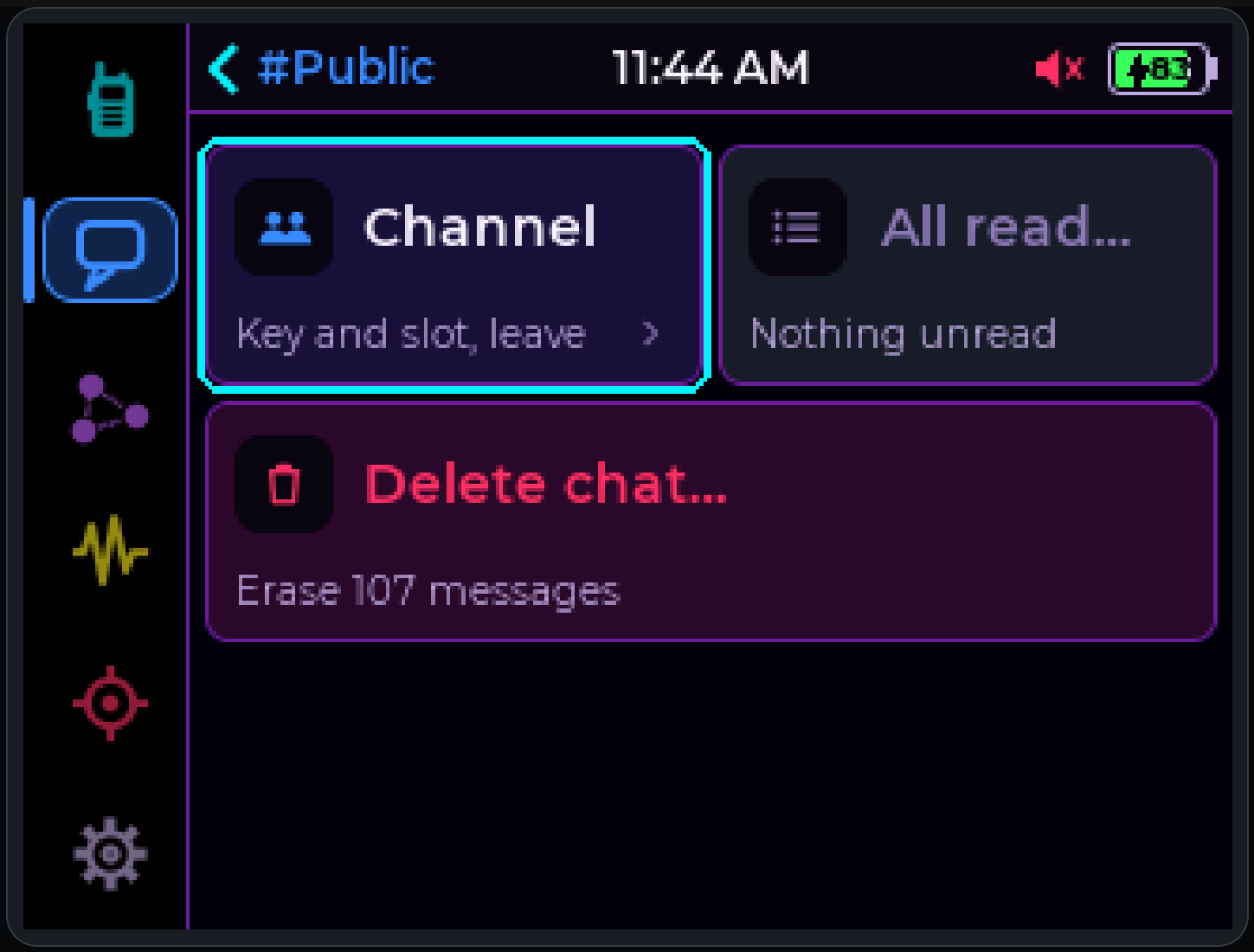
M9 A delivered message: no Resend card.

The chat's ⋯ page

Tap ⋯ at the top right of a thread. A channel shows **Channel**, **All read...** and **Delete chat...**. A direct chat with a saved contact adds:

- **Reset path**, with the stored path as its summary (**via A1 › 3F, Direct · in range**). A tap forgets it: **Path reset · next message floods.** The next reply learns a fresh route.
- **Find path**: floods a request for a route and stores the one the answer took (**Path found · 2 hops**).

Delete chat... removes every message of the conversation from this device. Deleting a channel's chat keeps you in the channel.



M9 A channel's ⋯ page.

TIP

When direct messages to one contact keep failing, the stored path is usually stale: ⋯ › **Reset path**, then send again.

Unread and new-message alerts

A thread you open is marked read. Unread shows as the Chats rail badge, the Home card's pill and, on text-entry pages where the rail is hidden, a red envelope in the status bar.

When a message arrives while you are using the device, a notice appears in the status bar; tap it to open the thread. When the device has been idle with **Msg wake** on, the screen wakes and shows a full-screen preview with **Dismiss**, **Next ›**, **Open** and **All read**. A screen you turned off yourself stays dark; the message waits. Message tones are set in **Settings › Hardware › Sound** and don't play while a phone app is connected.

History and sync

- The device keeps up to **200 messages** across all conversations; the oldest go first.
- **Keep chats** (**Settings › System › Restart**, on by default) saves them across restarts, unencrypted. Turning it off erases the saved copy.
- Messages a phone app sends appear in the device's chats as yours, with the delivered tick when confirmed. With CairnOS, messages sent on the device appear in the app.

09 / CHANNELS AND SCOPE

Who hears what, and where.

Channels decide who can read a message. Scope decides which regional repeaters carry it.

Channel kinds

Public

The well-known key every MeshCore node starts with. Everyone on the mesh can read it.

Hashtag

The key is made from the name, **#** included. Anyone who knows the name can read it. Capitals count: **#Test** and **#test** differ.

Private

A random key shared by hand: 32 hex digits or base64. Only people with the key can read it.

Joining and leaving

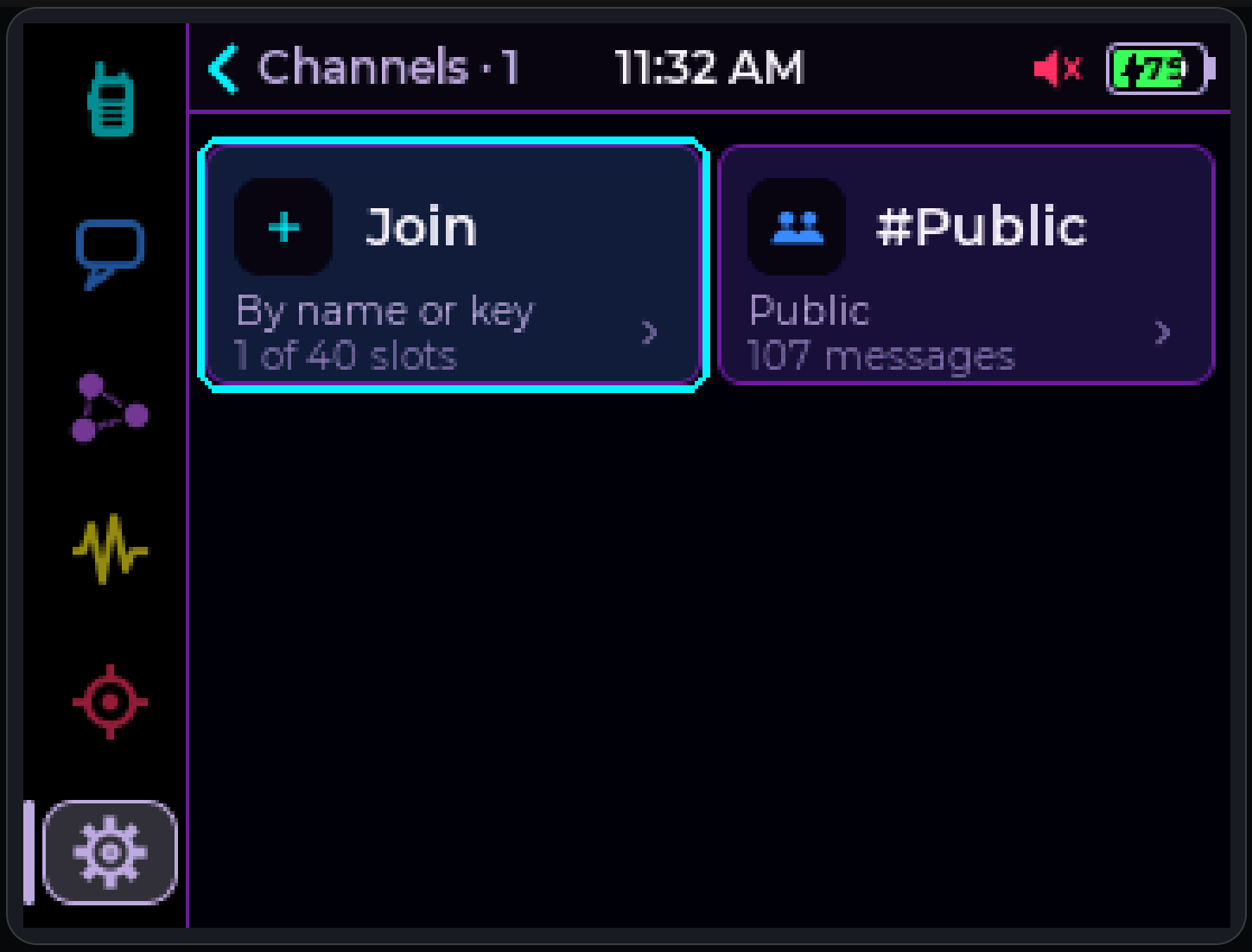
01 Open **Settings › Radio › Channels** and tap **Join** (it shows the slots used, up to 40).

02 Choose **Hashtag**, **Private** or **Public**.

03 Type the name, or the name and key. **New key** makes a random key for a new private channel.

04 Tap **Join**. **Joined #name** confirms it; nothing is sent over the air.

Tap a channel for its page: kind and slot, **Open chat**, **Scope**, the key (a private key stays hidden until tapped), and **Leave**, a hold-to-confirm tile below the first screen. Stored messages stay in Chats after you leave. A private key is not kept, so you need it again to rejoin.

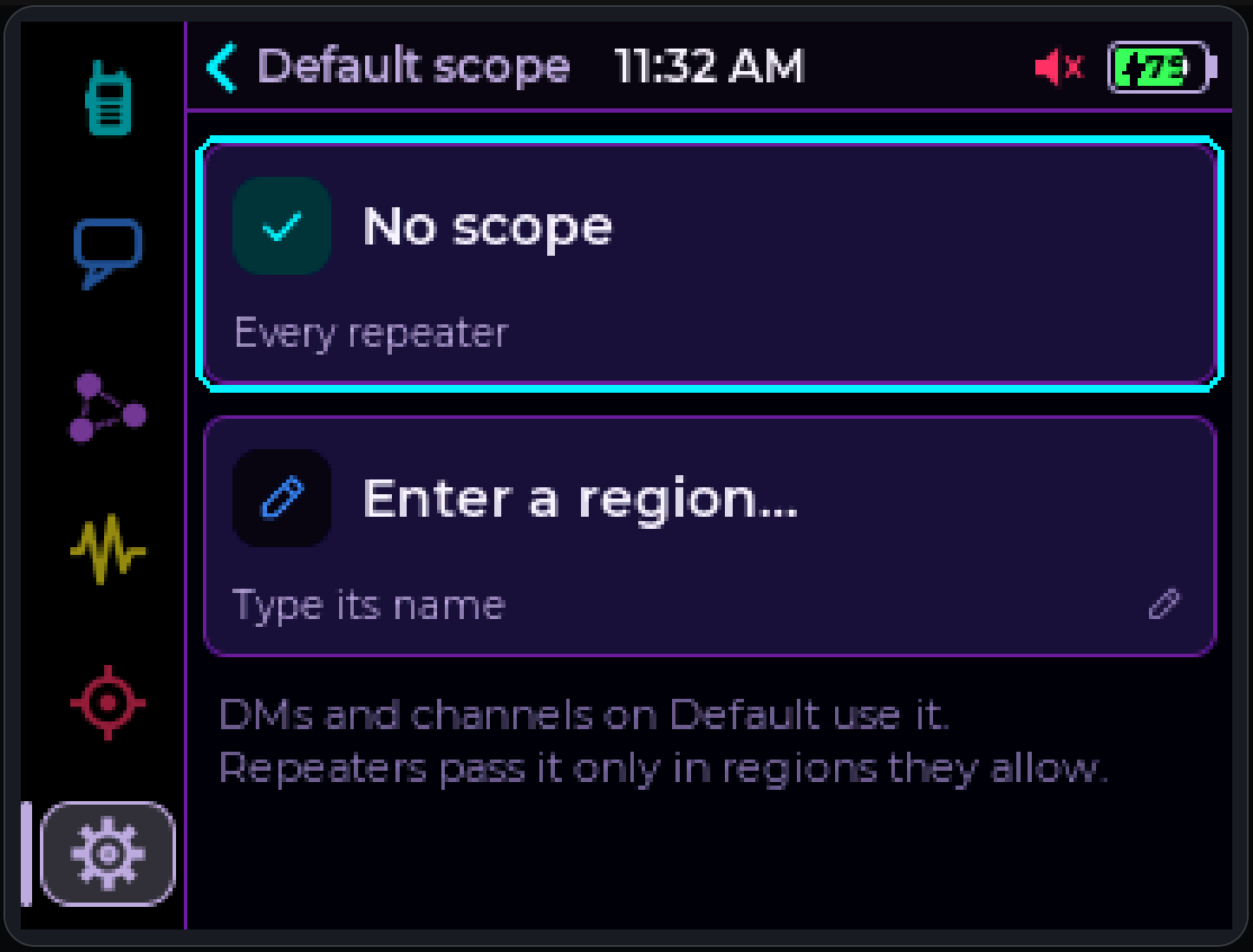


M9 Channels as cards.

Scope (regions) 118

Some meshes run regional repeaters. A repeater set to a region only passes floods *scoped* to a region it allows. The phone apps scope each message they send. From Cairn 118 the device does too.

- **Default scope:** **Settings › Radio › Scope**: **No scope**, a region you have used, or **Enter a region…**. Type the name as your repeaters use it, without the **#**; capitals matter, so a name with capitals asks first. This is the same default the phone apps set and show: change it in either place and both follow.
- **Per channel:** a channel's page has a **Scope** tile: **Use default**, **No scope** or a region. It is kept by the channel's key, so a channel you leave and rejoin starts on the default again.



M9 **Settings › Radio › Scope**.

YOU SEND	FROM	SCOPE USED
Channel message	the device	the channel's own scope, else the default
Direct message	the device	the default (a message with a known path goes direct, unscoped)
Any message	the phone app	the app's choice; a channel with no app scope uses its device scope

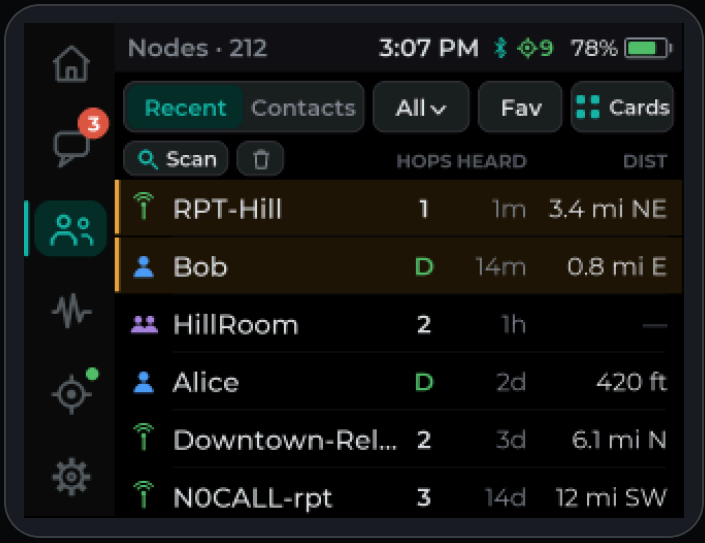
The pages are on the L2 Pro, Heltec kit, M9 and T-Deck. On the T-Display SF32 and the Wio L1 family, messages typed on the device use the default set from the app. Private (`$`) regions can't be entered on the device.

TIP

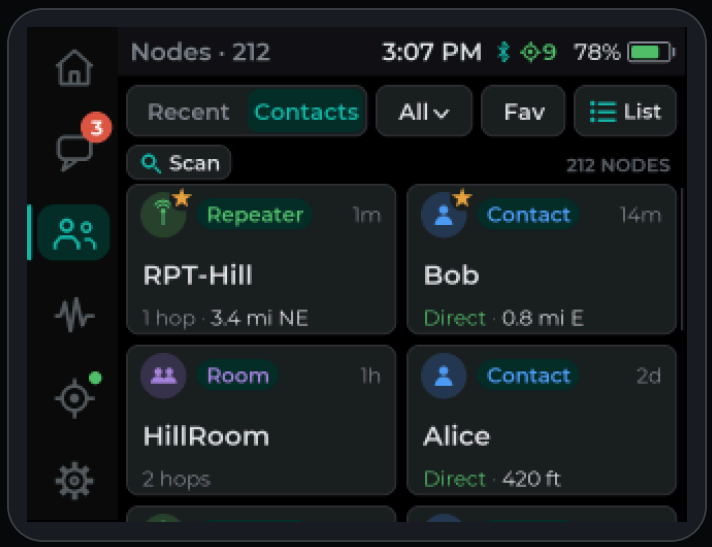
If phone messages reach far but messages typed on the device stop at the first hop, your mesh probably uses regions. Set the same default scope the phone app uses.

Who is out there.

Recently heard adverts and your saved contacts, as a list or as cards. Ping them, message them, map them, and log in to repeaters and room servers.



L2 PRO · EARLIER BUILD List view, sample data. Since Cairn 107 the Scan and trash chips are gone (see below).



L2 PRO · EARLIER BUILD Cards view, sample data. Today's cards add the short ID and position.

The toolbar

- **Recent** / **Contacts**: recently heard adverts (including nodes you haven't saved) or your saved contacts. The device holds up to 350 contacts; the title warns when the table is nearly full.
- Filter: **All**, **Contact**, **Repeater**, **Room**. In Recent, **Clear recents...** is the menu's last item.
- Sort: **Heard**, **Name**, **Distance**, **Favorites first**.
- **Cards** / **List** switches the view.

The title counts them: **Nodes · 9**, or **Nodes · 4 of 9** when filtered. Companions show a walkie-talkie, repeaters a mast, rooms a group. Favourites get an amber tint (list) or star (cards). List rows show **HOPS** (**D** for direct), **HEARD** and **DIST** with a bearing; distance needs your own GPS fix and the node's shared position.

TOUCH

- TAP a node for its detail page.
- HOLD it for its actions: Ping, Message, Details, Contact (or Admin on a repeater).

- PULL DOWN at the top of the list until **Release to scan**: a repeater scan.

KEYS

- OK opens, Hold OK for actions.
- ↑ past the top of the list scans for repeaters.
- In Cards view the ring takes one card at a time; ← → move between the two in a row.

TIP

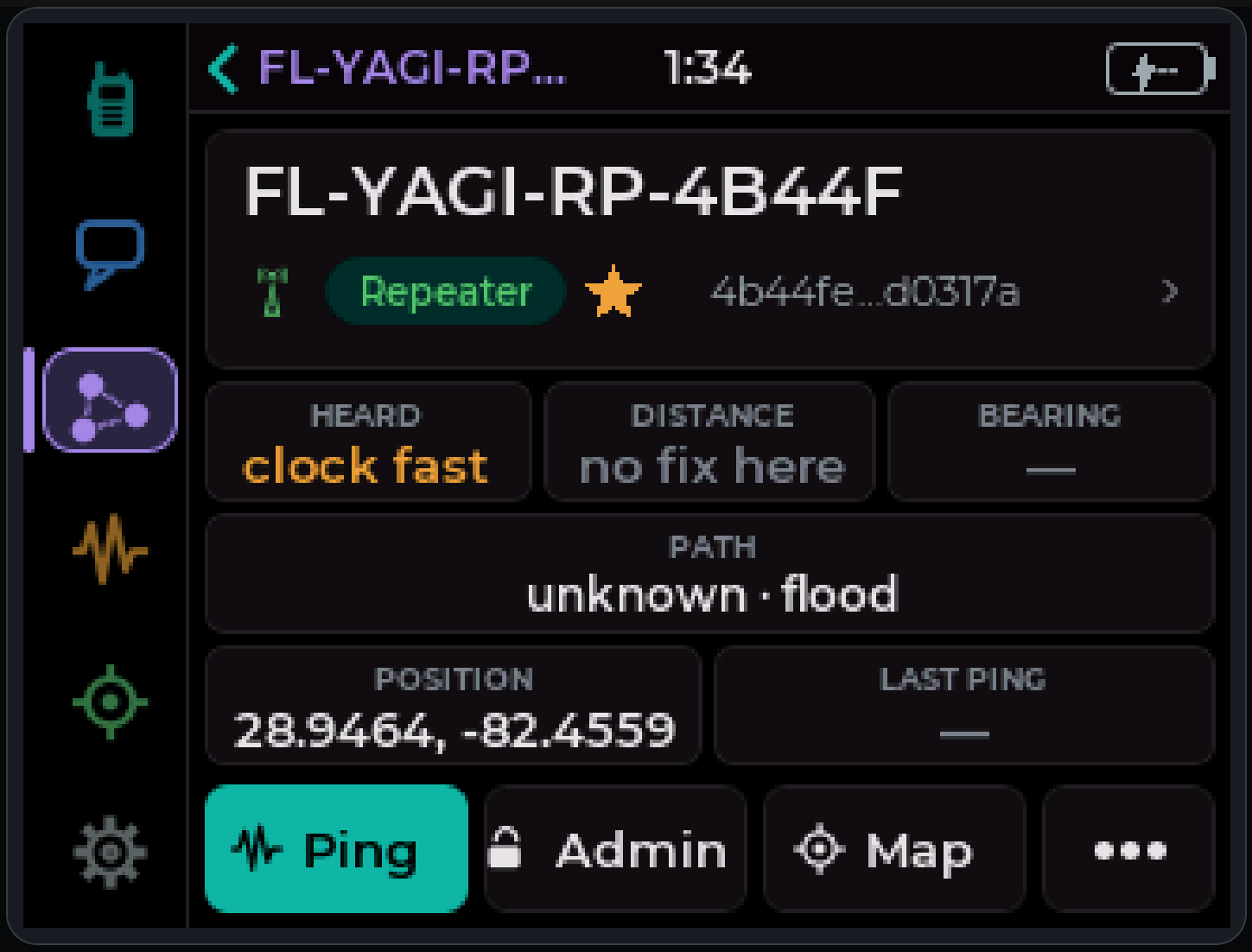
When the title warns the contact table is nearly full or full, tap the title: a notice says what that means for new nodes. Favourites are never overwritten.

Node detail

The name, then a row with the type, the favourite star and the short key (**4b44fe…d0317a ›**). Cells show **HEARD**, **DISTANCE**, **BEARING**, **PATH** (**direct**, **1 hop via 4e**, **unknown · flood**), **POSITION** and **LAST PING**. A heard time marked **clock fast** means the node's own clock is ahead of yours.

Buttons: **Ping**, **Chat** (or **Admin** on a repeater), **Map** and ⋯.

- **Ping** checks the node can hear you: **RTT 123ms SNR 7.5dB**, or **Ping timeout**. One at a time.
- **Map** opens the GPS map centred on the node, with you both in view.
- **The short key** opens a QR code the MeshCore apps can scan to add the contact, with the full key under it.



T-DECK · 113 A repeater's detail page.

Find path

Flood a request; store the route the answer took

Reset route

The next message floods and learns a new route

Favorite

Pinned in sorts; never overwritten when full

Share contact

Its advert goes out zero-hop so nodes in range add it

Admin / Log in

Repeaters and room servers

Remove contact...

Hold 1.5 s, then confirm

Above: the node's ••• page. Removing a contact keeps its messages in Chats; its next advert may add it back, depending on your auto-add setting in the app.

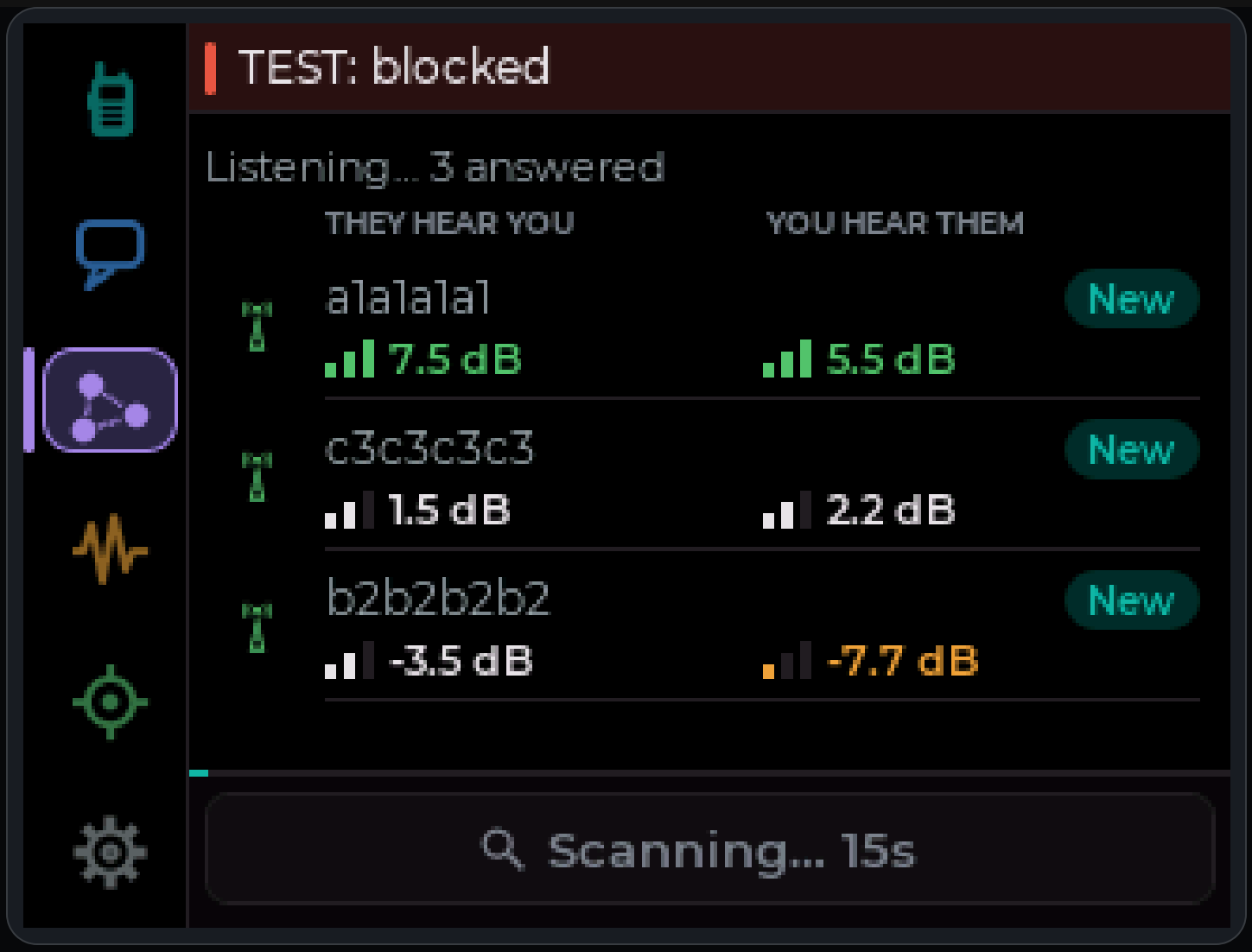


T-DECK · 113 Your own key's QR sheet, from Settings › System › About.

Repeater scan

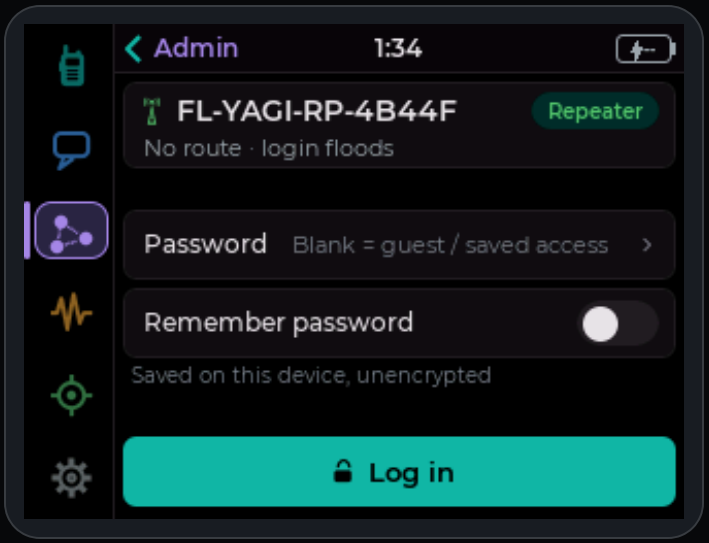
Finds the repeaters in direct radio range and shows how well each direction works. Start it by pulling down on Nodes (keys: ↑ past the top), or with the **Scan repeaters** button action, then **Scan for repeaters**. It listens for 15 to 45 seconds.

Each row shows the repeater, a **Contact**, **Heard** or **New** chip, and signal both ways: **THEY HEAR YOU** and **YOU HEAR THEM**. Tap one you haven't saved for **Add as contact** or **Add and log in…**. A 30-second cooldown follows each scan.

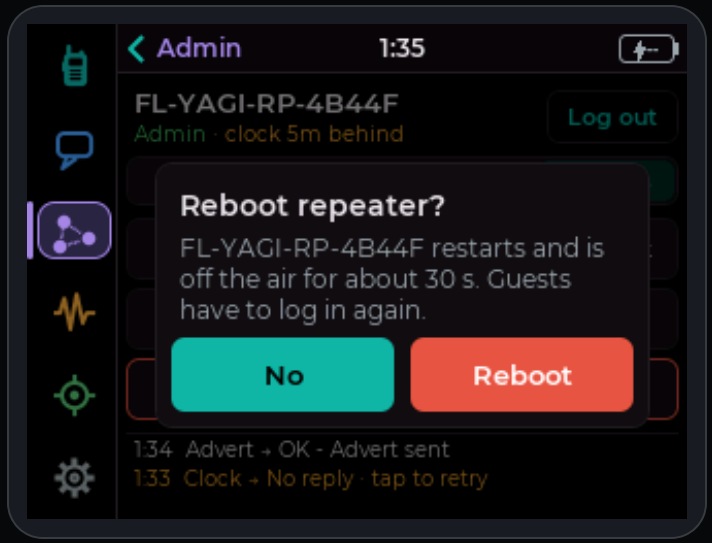


T-DECK · 113 **Scanning... 15s**.

Repeater and room admin



T-DECK · 113 Log in. Blank is guest access.



T-DECK · 113 Actions › Hold to reboot asks once more.

01 Open a saved repeater's detail page and tap **Admin** (a room: **Log in…**). The device clock must be set.

02 Type the password (up to 15 characters) or leave it blank for guest access. **Remember password** saves it on this device, unencrypted.

03 Tap **Log in**. A repeater never says "wrong password"; it just doesn't answer. If a direct login fails, **Try flood** resets the route and floods the login.

Logged in, the page has tabs: **Status** (battery, uptime, clock, noise floor, airtime, packet counts), **Nearby** (repeaters it hears, repeaters only), **Sensors** and **Actions** (admins only): **Advert**, **Flood advert**, **Sync clock**, **Radio**, **Clock**, **Version** and **Hold to reboot**. Requests are paced to save airtime: one at a time, with a short wait after each. There is no free-text command line.

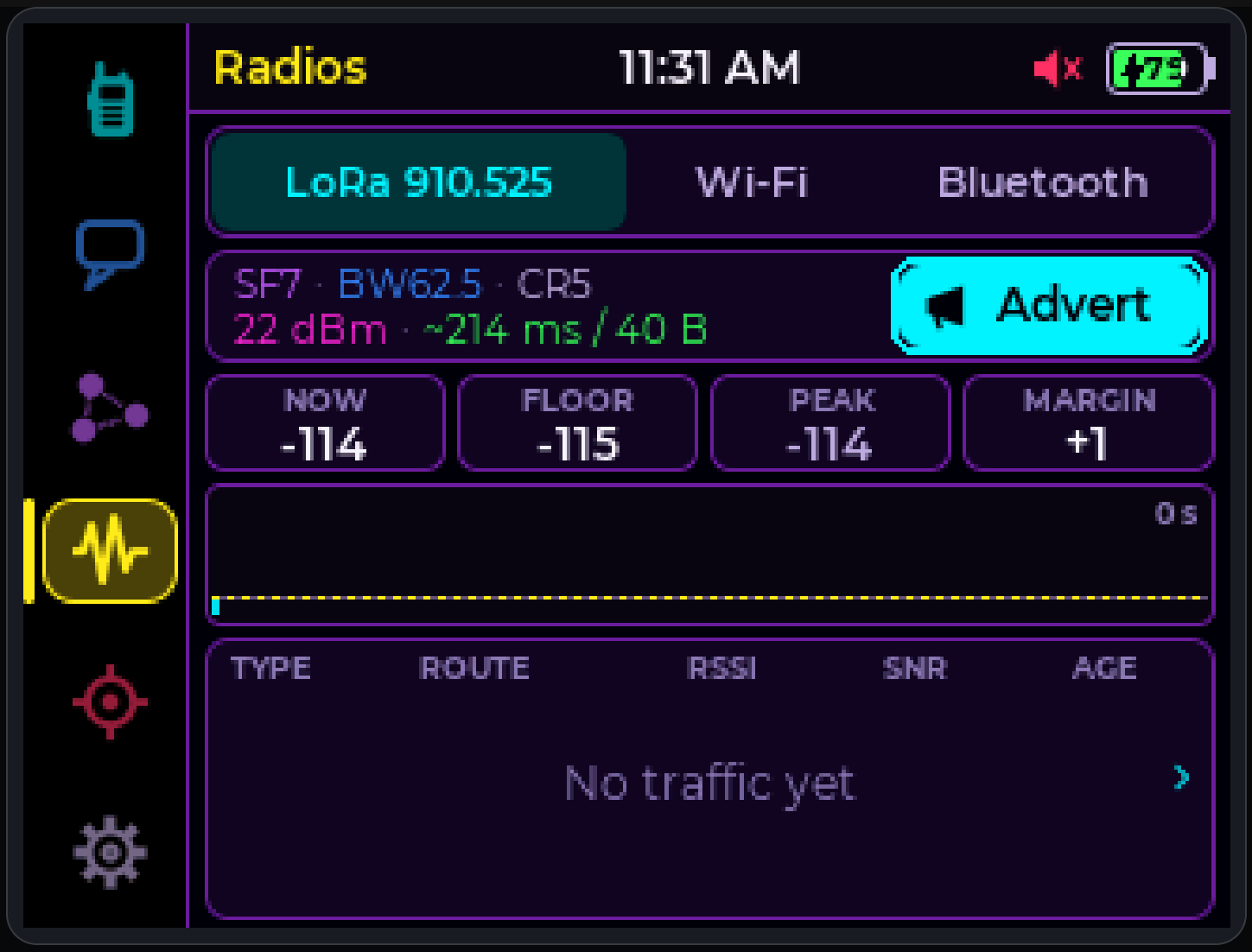
Rooms. Log in to a room server and it appears in Chats as **@Room**. Posts always appear to come from the room.

See the air.

One tab, three radios: **LoRa** (its tab shows your frequency), **Wi-Fi** and **Bluetooth**. The rail cell comes back to the tab you picked last.

The LoRa page

- **Header:** SF, bandwidth and coding rate over TX power and the airtime of a 40-byte packet (**~214 ms / 40 B**), with a large **Advert** button.
- **Meters: NOW** (latest reading), **FLOOR** (noise floor), **PEAK** and **MARGIN** (now over floor, green above 6 dB).
- **Graph:** about 80 seconds of signal. Tap it for **Spectrum**.
- **Traffic card:** the newest packets with **TYPE**, **ROUTE**, **RSSI**, **SNR** and **AGE**. Tap it for **Traffic**.



M9 By keys the ring lands on **Advert**.

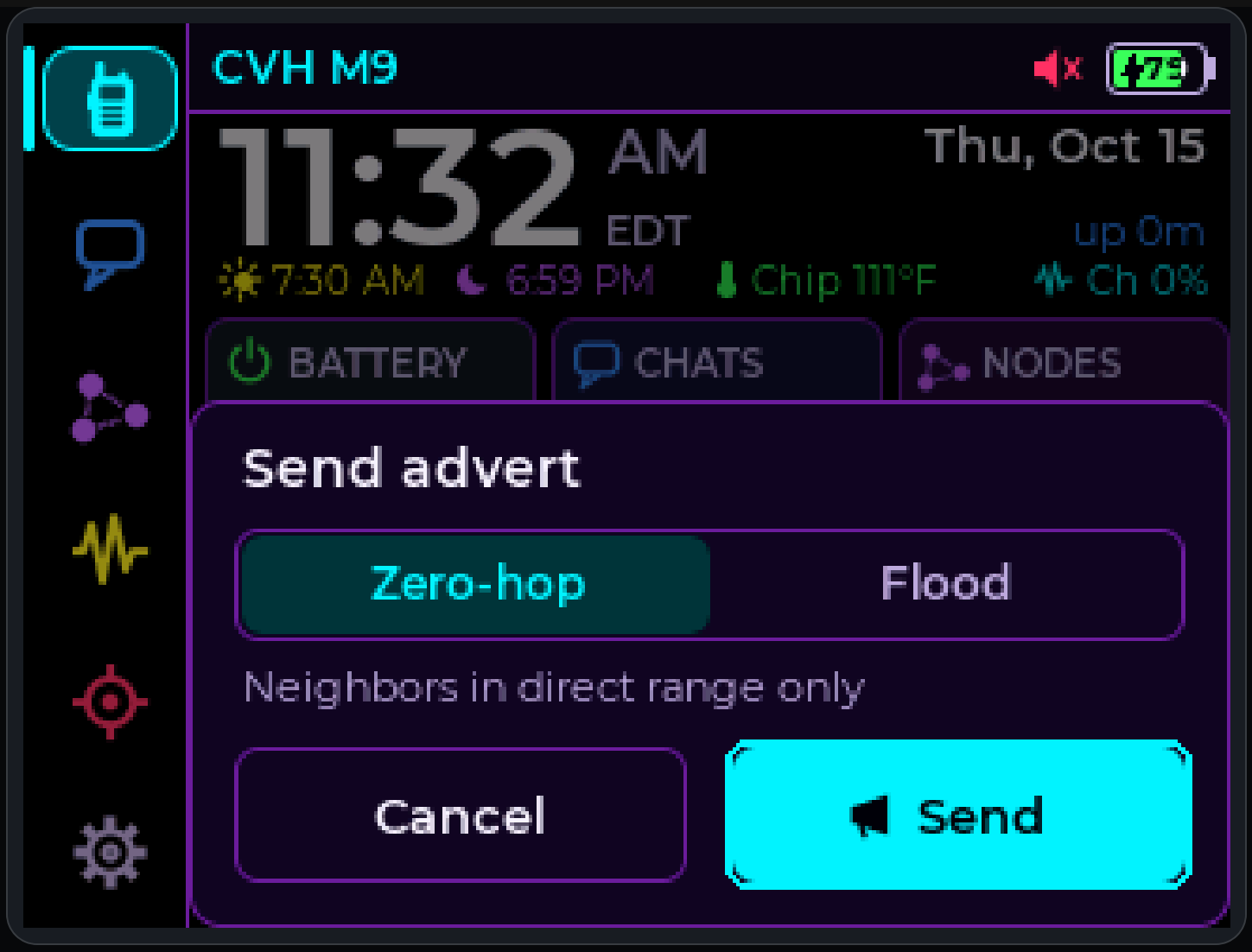
Sending an advert

An advert announces your node. Choose **Zero-hop** (**Neighbors in direct range only**) or **Flood** (relayed across the mesh, scoped). Nothing is sent until you tap **Send**. It carries your position if Profile's **Position** is on.

Open the sheet from the LoRa page's **Advert**, the quick panel's **Advert** tile, a hold on Home's LoRa card, the **Send advert** button action, or ADV on the M9.

TIP

By keys the sheet opens with **Send** selected and Zero-hop chosen: on the Radios tab, OK OK sends a zero-hop advert.



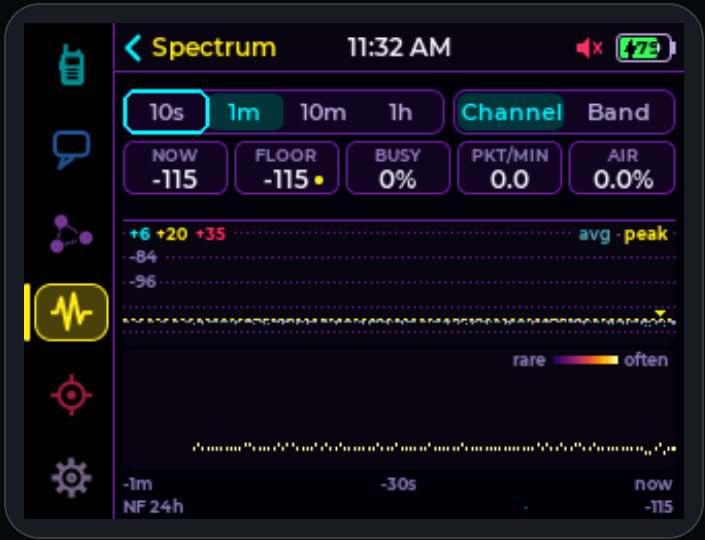
M9 Opened with ADV.

Traffic and packet details

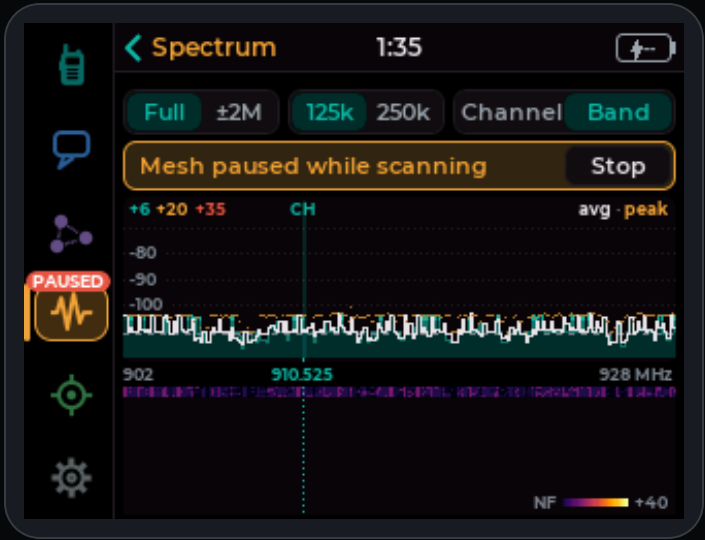
Traffic lists every packet heard over about the last hour, up to 2048, newest first, under a strip with **PACKETS**, **PER MIN**, **AVG RSSI** and **AVG SNR**. Tap a packet for its window: the sender first, then type, route and hops, signal, margin over the spreading factor's limit, from (or channel), to, heard via, size with airtime, and when it was heard. **Open node** jumps to a saved sender's page.

CODE	PACKET	CODE	PACKET
MSG	direct text message	GTX	group or channel text
ACK	acknowledgement	GDA	group data
ADV	node advert	PTH	path return
REQ / RSP	request / response	TRC	trace
ANQ	anonymous request	MPT	multipart
CTL	control	RAW	raw or custom

Spectrum: Channel and Band



M9 Channel view: your channel over time.



T-DECK · 113 Band view: the mesh pauses while it sweeps.

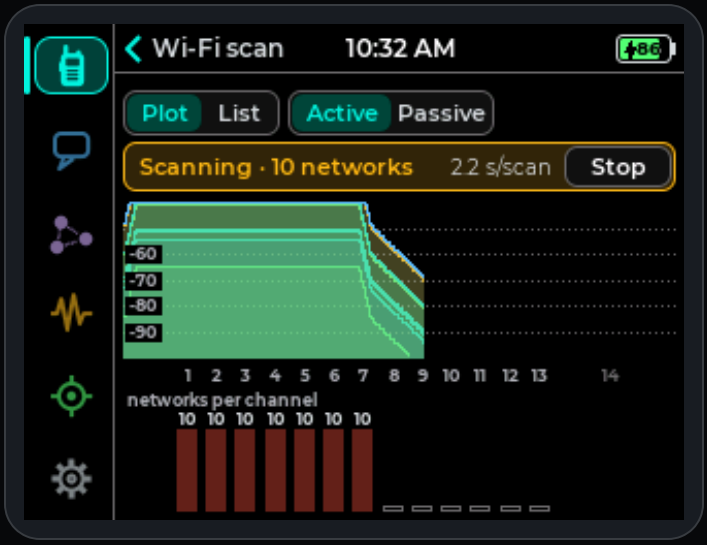
Channel only listens and never pauses the mesh. Pick **10s**, **1m**, **10m** or **1h**. The strip shows **NOW**, **FLOOR**, **BUSY** (amber above 30 %, red above 60 %), **PKT/MIN** and **AIR**. Under the trace a density map shows how often the signal sat at each level, and **NF 24h** traces the noise floor over a day. Tap a packet marker for its details, or the trace for a readout at that moment.

Band sweeps the whole band (**Full**) or ±2 MHz around your channel, at **125k** or **250k** steps, with a waterfall. Tap **Start**: the banner turns amber, **Mesh paused while scanning**, and the rail shows **PAUSED**. Your device sends and receives nothing until you tap **Stop**. It also stops

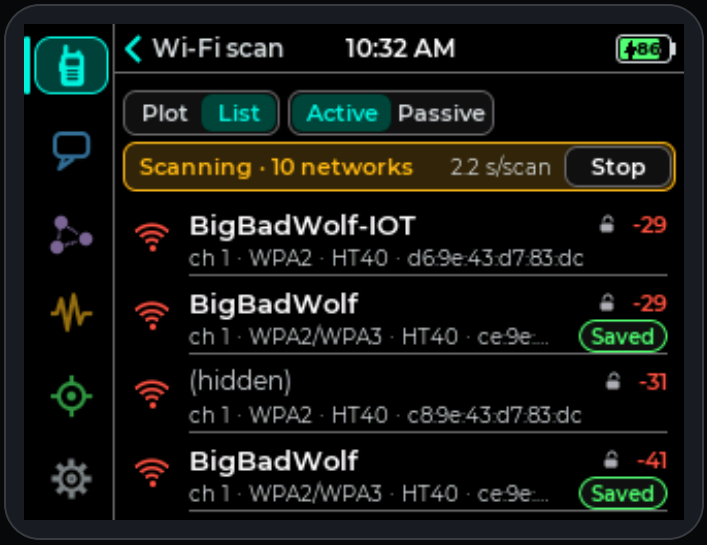
when you leave the page, the screen goes dark, after 3 minutes without a touch, or when the mesh needs to send.

> TIP
> Hold the band plot to clear the peak hold. Tap the plot or a waterfall row for a frequency and level readout; tap the same spot again to clear it.

Wi-Fi scan



L2 PRO · 113 Plot. From 116 the controls sit at the bottom.



L2 PRO · 113 List, strongest first.

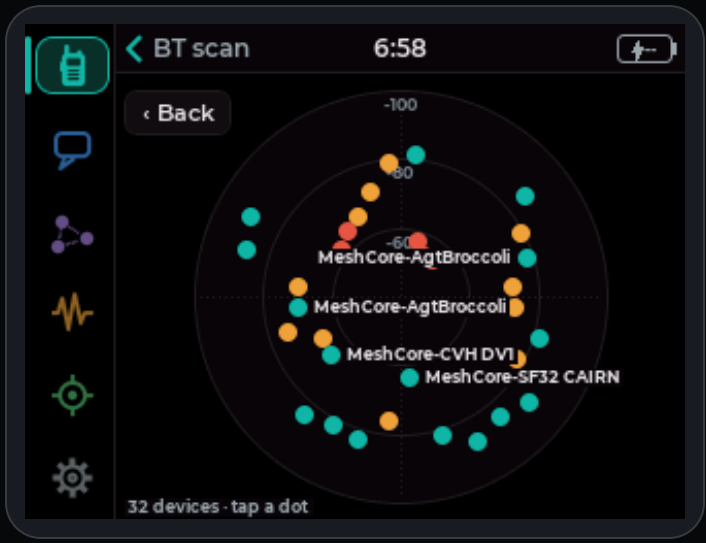
Shows the 2.4 GHz band, channels 1 to 14: one arc per network at its signal strength, the four strongest named, your own network's channel highlighted, and a bar per channel for how crowded it is. Tap the plot or a bar for a readout. The list names every network, hidden ones too, with channel, security and a **Connected** or **Saved** tag; tap one for **Details**, **Show on plot** or **Join…**.

Active asks for names (probe requests); **Passive** only listens. A scan starts only when you tap **Start**, and stops when you leave the page, the screen goes dark or after 3 minutes without a touch. Nothing is sent on the mesh and LoRa keeps working. The status bar shows **Scanning** and the count.

Bluetooth scan, radar and Find



T-DECK · 113 Bars over a half radar.



T-DECK · 113 Full-screen radar: distance only.



T-DECK · 113 Find: walk until it says Closer.

Shows the Bluetooth LE devices in range as bars over a half radar with -60, -80 and -100 dBm rings; MeshCore nodes are named in the accent colour. Tap the radar for a full-screen view: it shows distance only, never direction. Pick a device and **Find** shows its signal in big figures, one word for the last 10 s (**Closer**, **Farther**, **Steady** or **Lost**) and the last minute as a line; the optional **Tone** ticks faster as the signal climbs. BT scan works with Bluetooth off and while the app is connected.

TIP

On the full-screen radar the d-pad or trackball jumps to the nearest dot in the direction you press. On the T-Deck, keys `1` `2` `3` pick the LoRa, Wi-Fi and Bluetooth tabs, and `#` switches Plot and List.

Where you are, and who is near.

Position, course and speed, a GPS data page, and a map with you and every contact that shares a position. With a prepared microSD card, the map draws real tiles, fully offline.

GPS modes

Off

The receiver is powered down. Position and the map keep the last fix.

On

Tracks continuously. The biggest single drain on the battery.

Eco

Wakes every 10 minutes for a fix, gives up after 90 s, then sleeps again.

Set the mode in **Settings › Location › GPS**, with the switch at the top right of the GPS page, the quick panel's GPS tile, the **Toggle GPS** button action, or a hold on `ADV` (M9). The rail's GPS icon is green while on and red while off.

The GPS page

Position shows **LAT**, **LON**, **ALT**, **SPEED** and **FIX**, a status pill (**GNSS · 9 sats**, **SEARCHING**, **NO GPS DATA**, **ECO · next 3:40 PM**) and a compass. The compass shows your course over ground, so it only points while you move.

Tap the status or any card for **GPS data**: satellites used and in view, HDOP and a rough accuracy, position, altitude, speed, course, the Maidenhead grid square, GPS time, the last fix and the raw NMEA stream.

Units follow **Settings › Location › Units**: Imperial or Metric.

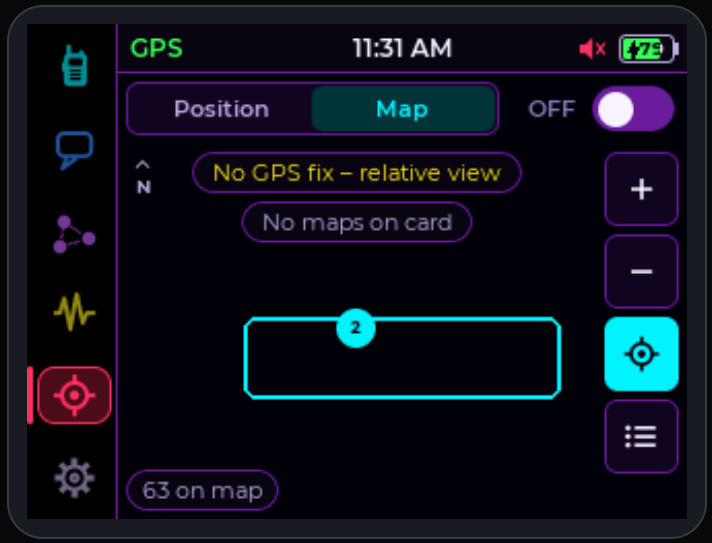


L2 PRO · EARLIER BUILD Position, sample data. Today an on/off switch sits at the top right.

The map



L2 PRO · EARLIER BUILD Radar view with nodes, sample data.



M9 Status chips explain what you see.

North is up. Where a map set covers the view you see tiles; elsewhere a radar view with range rings around you. Nodes are coloured dots by type; nearby ones merge into a bubble with a count. Your position is a filled dot during a live fix, an arrowhead while you move, a hollow ring for the last known fix. The map remembers your last fix across restarts.

TOUCH

- DRAG pans; PINCH or double tap zooms.
- Tap a node: its card and its route, drawn through each repeater with a position. Tap the card for its page.
- Tap a bubble to zoom in on it.
- HOLD empty map: **Map card info**, **Choose map...**, **Reload card**.
- Tap the GPS tab again to recentre.

KEYS

- + / - zoom.
- M9 MAP opens the map; on the map, MAP again is pan mode: arrows pan, OK taps the centre.
- The buttons on the right (zoom, follow, list) take the ring.

BUTTONS ON THE RIGHT

- + / –

- **Follow**: recentre on you and follow.
- **List**: Nodes sorted by distance, for that visit.

Chips at the top explain the view: **No GPS fix – relative view**, **No map card**, **No maps on card**, **Zoom in for map**, **No map for this area**, **Card isn't FAT32**. Tap a card chip to retry. The attribution in the corner opens the map's licence.

> TIP
> A node page's **Map** button centres the map on that node and fits you both in view. Tap the GPS tab again to come back to your own position.

Choosing a map set

Settings › Location › Map lists **Automatic** and each set on the card with its zoom range (**okc-topo · z8–16**). **Automatic** uses the most detailed set that covers the view. A set you pick is used wherever it covers the view, and Automatic elsewhere. The choice saves at once.

Building map sets with tools/maptiles.py

The device never downloads maps. You build a set on a computer and copy it to the card. USGS topo is the default source: public domain, no key, US only, up to zoom 16.

01 **Set up once** (Python 3.9+): `python3 -m venv ~/.venvs/l2tiles && ~/.venvs/l2tiles/bin/pip install Pillow numpy`, then `python3 tools/maptiles.py selftest`.

02 **Prepare a FAT32 card.** 8 to 32 GB SDHC is best; exFAT isn't read. On a Mac: `diskutil eraseDisk FAT32 CAIRN MBRFormat /dev/diskN` (erases the card), or format it on the device under **Settings › Hardware › Storage**.

03 **Estimate, then build.**

```
python3 tools/maptiles.py estimate --center 35.0,-97.0 --radius-mi 10 --zoom 8-16
python3 tools/maptiles.py build --preset usgs:topo --center 35.0,-97.0 --radius-mi 10 \
    --zoom 8-16 --out /Volumes/CAIRN --name okc-topo --title "OKC topo"
python3 tools/maptiles.py verify /Volumes/CAIRN/Maps/okc-topo
```

04 **Insert the card.** The device notices a new set by itself.

AREA (USGS TOPO)	ZOOMS	CARD SIZE	FIRST DOWNLOAD
±5 mi	8–16	~33 MB	~13 min
±10 mi	8–16	~126 MB	~50 min
±25 mi	8–15	~200 MB	~1 h 20 min
±100 mi	6–13	~195 MB	~1 h 15 min

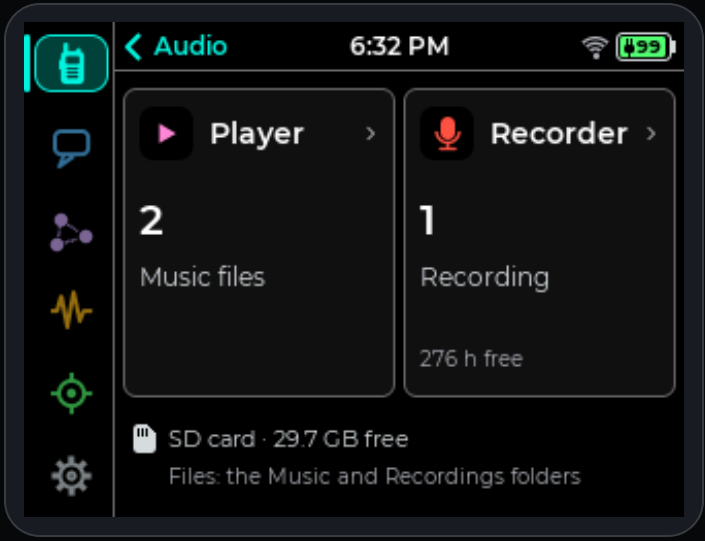
- One screen is about 0.4 mi across at zoom 16, 1.6 mi at 14, 6 mi at 12 and 25 mi at 10. Each extra zoom level is about four times the tiles. A good plan: one wide, shallow set plus a deep one around home.
- Up to 8 sets per card, each in `/Maps/<name>/`. The device magnifies the deepest level up to 2 more zooms; it looks blocky.
- Other sources: Geoapify (free key), Thunderforest (bulk plans only), your own MBTiles, or a local tile server. OpenStreetMap's tile server, OpenTopoMap, CARTO, Esri, Google, Bing, MapTiler Cloud and Stadia are refused: their terms forbid bulk or offline use.
- Filters: `--filter night` (a dark map), `dim:N`, `gray`.
- A card you build is for your own use. Don't share or sell it.

FOLDERS

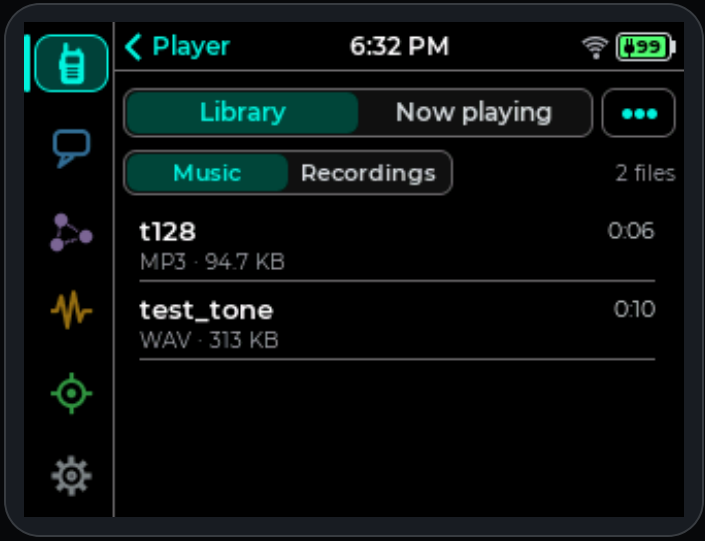
Older firmware read maps from `cairn/maps` (and before that `l2map`). Current firmware moves `cairn/maps` sets to `/Maps` by itself the first time it reads the card; nothing is copied or deleted.

Music, voice notes, your own alerts.

On the Wio Tracker L2 Pro and the T-Deck (both have a speaker and a microphone). The Heltec kit and the ThinkNode M9 have only a buzzer, so their builds have no player.



L2 PRO · 113 The Audio hub.



L2 PRO · 113 The Library.



L2 PRO · 113 The Recorder, with the **MIC** pill live.

Player

Put MP3 or WAV files in a `Music` folder on the card; subfolders show two levels deep (`Music/Artist/Album`). Open the player from Home's Audio card (one swipe down), the

quick panel's **Audio** button or a button action. The **Library** has **Music** and **Recordings**; tap a file to play it, hold it to delete it.

Now playing fills the screen with a 24-bar visualizer, the level, the peak and the loudest frequency, the EQ in use and the time. Tap it and the controls slide in: title, Library, Options, volume, progress and previous / −10 s / play-pause / +10 s / next. Tap the bars to put them away.

Options (⋯) are cards: Repeat, Shuffle, Volume, EQ, Screen off and Alerts. The **EQ** has five bands (150 Hz, 400 Hz, 1 kHz, 2.5 kHz, 6 kHz, ±6 dB) with Flat, Voice, Bright, Warm and Custom; drag a band's handle over the live spectrum to shape Custom. Music keeps playing with the screen dark unless you choose otherwise. Alert tones don't interrupt music unless **Alerts** is on.

Plays MP3 (CBR or VBR) and WAV (8, 16 or 24-bit, mono or stereo), 8 to 48 kHz; stereo plays as mono. On the T-Deck the card shares a bus with the screen and radio, so very large files are refused with **Too fast for this card: use MP3**.

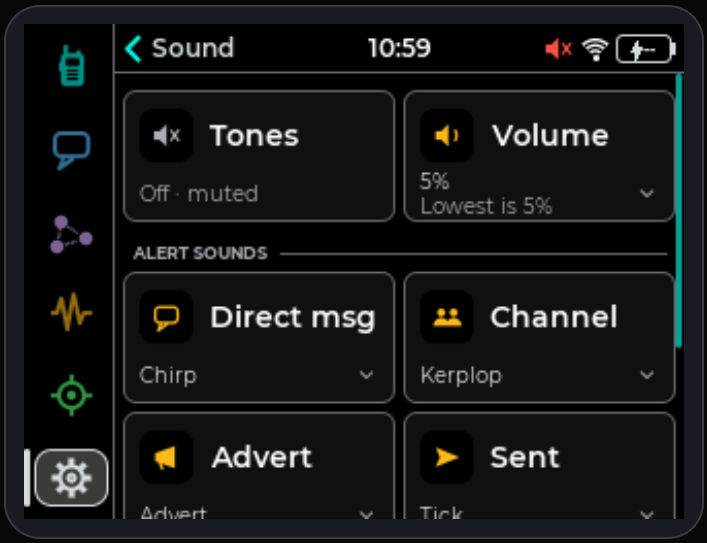
> TIP
> On the T-Deck keyboard: `Space` play/pause, `n` / `p` next and previous, `s` shuffle, `r` repeat, `#` switches view, `*` options. On Now playing, `←` `→` seek 10 s, `↑` `↓` change the volume and `0` – `9` jump to 0–90 %.

Recorder

Records voice notes to `Recordings` as WAV at 16 or 8 kHz, named by date and time once the clock is set. The gear sets quality, mic gain, a time limit and **Voice start** (it waits for sound and keeps the half second before it). **Mark** drops a marker. A recording keeps going with the screen dark and is kept safe if the power or the card goes.

The microphone is only on in the Record view, while voice start waits, and while recording; the status bar shows **MIC** or **REC** whenever it is. On the T-Deck the `Mic` key opens the recorder, and on the recorder starts and stops a recording (`r` or `Enter` too; `m` marks).

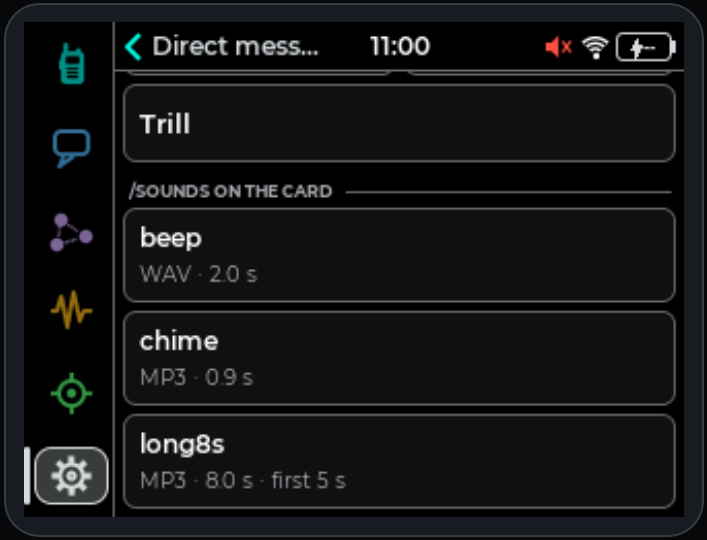
Alert sounds from the card 118



T-DECK **Settings › Hardware › Sound**.



T-DECK Built-in tones first.



T-DECK Then your files, with their length.

01 Copy MP3 or WAV files into a `Sounds` folder at the top of the card (a card reader, USB drive mode, or Wi-Fi transfer's **Sounds** target).

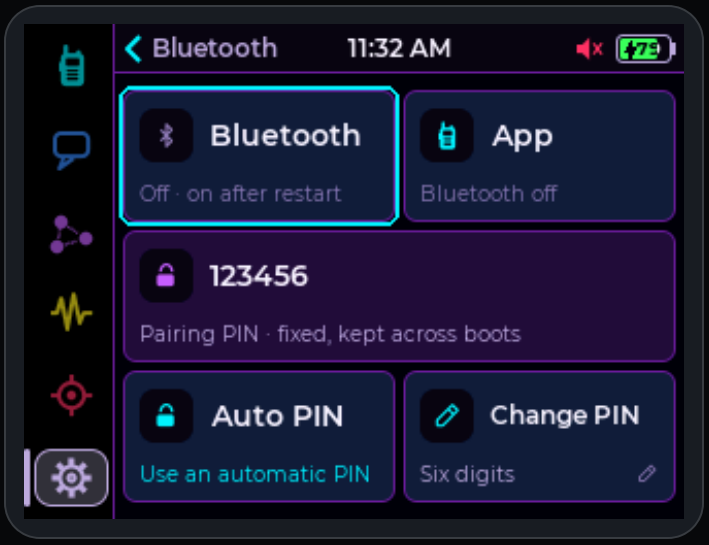
02 Open **Settings › Hardware › Sound** and tap **Direct msg**, **Channel**, **Advert**, **Sent** or **Startup**.

03 Scroll past **BUILT-IN** to **/SOUNDS ON THE CARD**. Each tap picks and plays. Back saves.

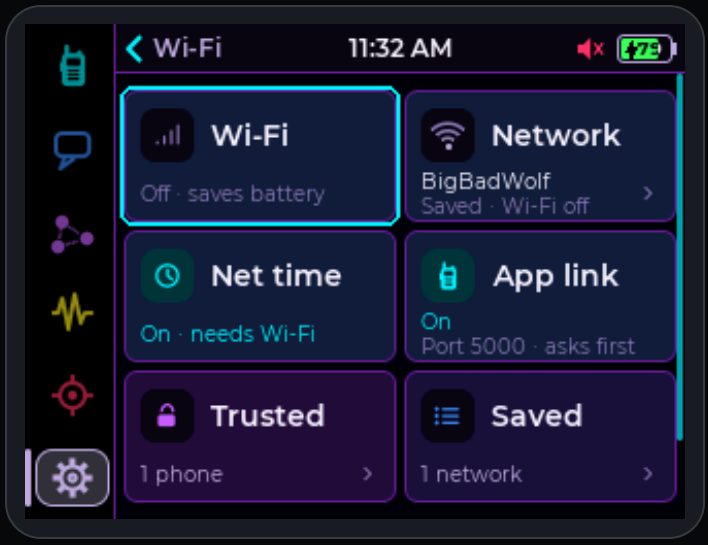
An alert plays at most the first 5 seconds of a file, fading out. The built-in tone plays instead while music is playing or paused, while the mic is on, or when the card or file isn't there. Mute silences both. **Reset UI** puts the tones back.

Link a phone. Move files.

Cairn works with the official MeshCore apps over Bluetooth, and on the four touch and key boards over Wi-Fi too. The card holds maps, music, recordings, sounds, pictures and screenshots.



M9 **Settings › Connect › Bluetooth**.



M9 **Settings › Connect › Wi-Fi**.

Bluetooth and the phone app

In the MeshCore app, connect to the device and type the pairing PIN shown on Home's Network card or on the Bluetooth page. The PIN is **auto · new each boot** unless you set one with **Change PIN** (six digits; **000000** goes back to automatic). **Auto PIN** switches back too. Turning Bluetooth off while the app is linked asks first.

Wi-Fi

- **Network** opens the list of networks around you. Tap one: a saved network joins at once, an open one asks, a secured one asks for its password. **Hidden network…** at the bottom joins by name. Up to 8 networks are remembered; Wi-Fi reconnects to the strongest in range.
- **Net time** sets the clock from the internet. A change of more than an hour asks first unless **Always sync** is on in **Settings › System › Clock**.
- **App link** lets the MeshCore app connect over Wi-Fi on port 5000. The first time a phone connects the device asks **Allow app over Wi-Fi?**: **Cancel**, **Allow once** or **Always allow**. Allow only your own phone.

- **Trusted** lists phones that connect without asking (up to 4). Hold a phone's trash icon to forget it.

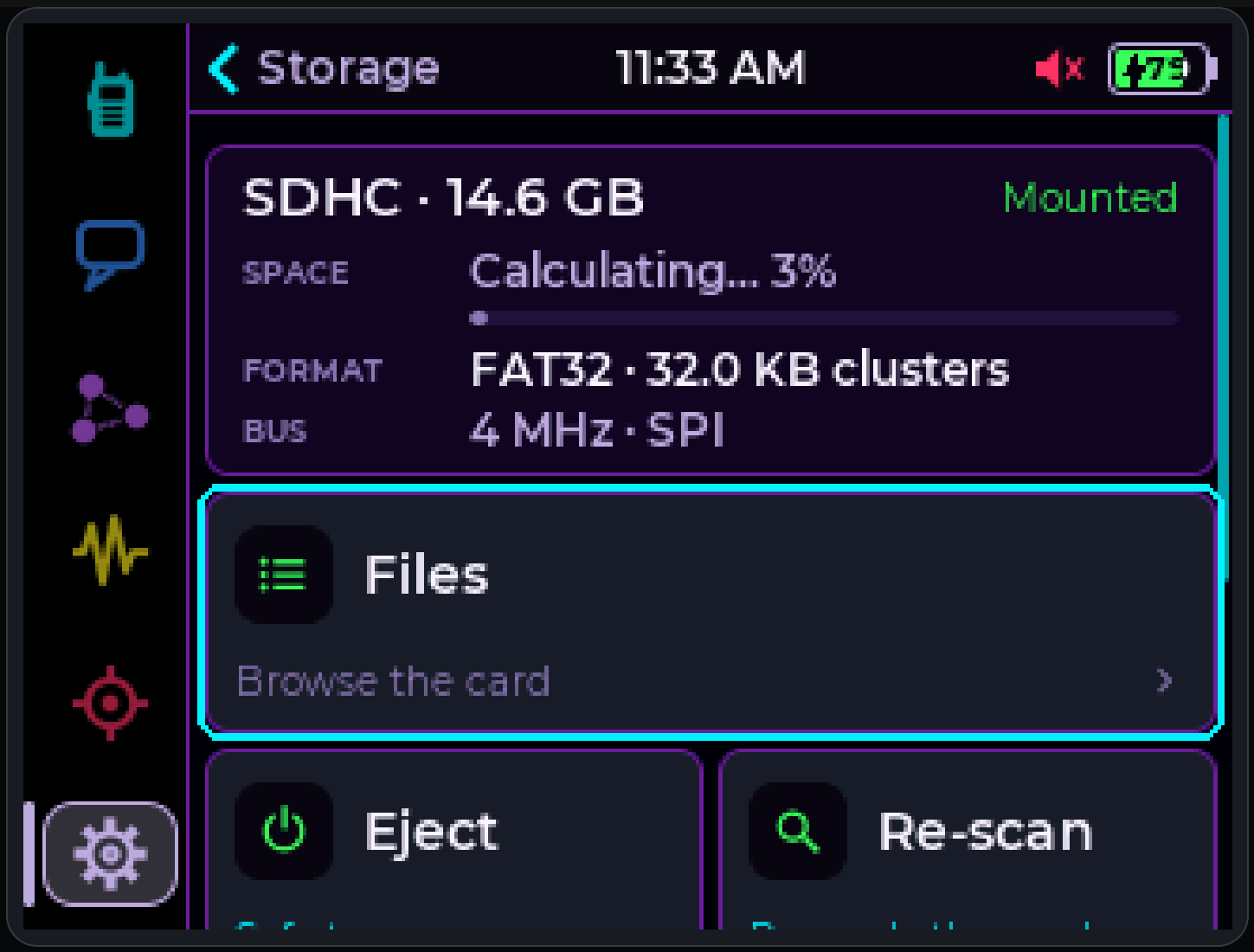
Wi-Fi adds about 20 to 30 mA while connected. Once connected, the Wi-Fi page shows the IP address.

> TIP
> Hold the quick panel's Wi-Fi or BT tile to open these pages straight away; ‹ returns you to where you were.

Storage and Files

Settings › Hardware › Storage shows the card (type, size, used and free, format, bus speed and whether it is mounted), then **USB drive** (L2 Pro and T-Deck), **Files**, **Eject**, **Re-scan** and **Format card** (hold 2 s, erases everything and makes the card FAT32).

Files browses the card a folder at a time, with sizes. Hold a file or folder to delete it; a folder delete runs in the background and survives a restart. Back goes up a folder. Tap a `.png`, `.jpg` or `.bmp` to view it full screen; swipe sideways (keys: `←` `→`) for the next one.



M9 Storage. The M9 has no USB drive mode.

Maps
Offline map sets

Music
Songs for the player

Recordings
The recorder's files

Sounds
Alert sounds (Settings › Sound)

Pictures
Images to view

Screenshots
From Capture and the Screenshot action

The same folder layout is used on every Cairn board, so a card moves between them.

USB drive mode

On the L2 Pro and T-Deck, **Storage › USB drive** turns the device into a USB disk: the card mounts on your Mac or PC and you copy files both ways (about 0.7 MB/s on the L2 Pro, 0.5

MB/s on the T-Deck). The radio and the app stay off meanwhile. Eject the disk on the computer, then tap **Done**: the device restarts and goes back to normal. Not on the Heltec kit or the M9, whose USB ports can't act as a drive.

Screenshots

The quick panel's **Capture** button takes the page under the panel; the **Screenshot** button action takes the current page. Both save PNGs to `/Screenshots`.

Every card, briefly.

Settings opens on six category cards. Each opens a page of cards, two across, each with its name and current value. A card that is on is washed in its colour and wears an **ON** pill.

Radio

USA/CAN · 22 dBm
910.525 · SF7/62.5

Connect

Wi-Fi off · BT off
No Wi-Fi network

Hardware

Bright 100% · Muted
Off 15s · Batt 79%

Location

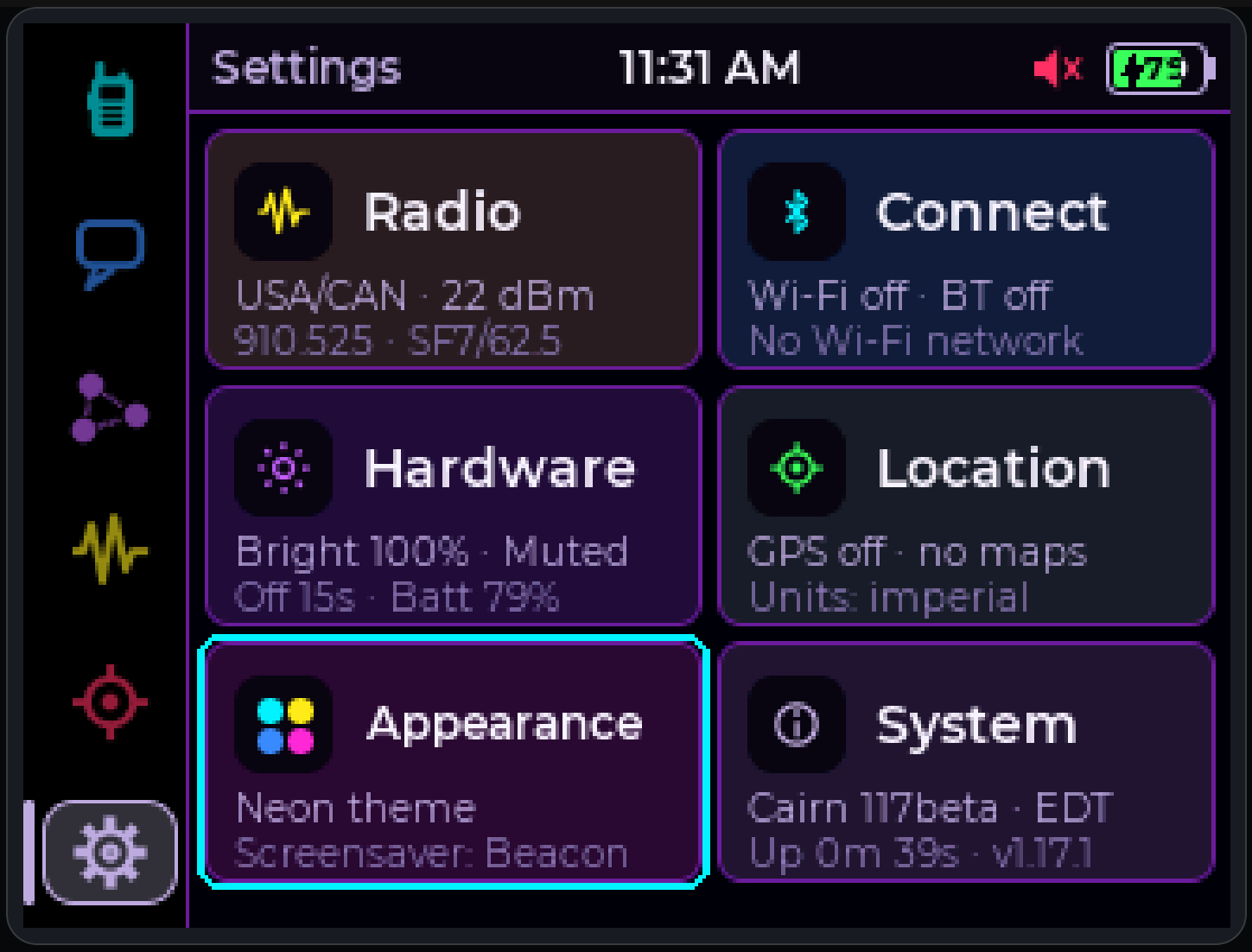
GPS off · no maps
Units: imperial

Appearance

Neon theme
Screensaver: Beacon

System

Cairn 117beta · EDT
Up 0m 39s · v1.17.1



M9 The six categories, as on the device.

Saving. Appearance, Display, Sound, Buttons, Wake & lock and Clock apply as you change them and save when you leave. Profile, Radio, the Bluetooth PIN and Join channel keep a draft with **Cancel** and **Save**; the title shows • **unsaved** and leaving asks **Discard changes?**. Switches act at once, asking first when they would drop a linked app.

TIP

Tap the Settings tab while deep in a page to jump back to the six categories. Cards with no page of their own, such as **Auto retry** and **Units**, flip with a single tap.

Radio

Profile

Node name (up to 31 bytes) and **Position** in adverts. With position on, adverts carry only 23 bytes of the name. Save offers to send an advert.

Channels

Join, open, scope and leave channels. Chapter 09.

Radio

TX power (with ~mW), then **Preset**, **Frequency** (typed, 150–960 MHz), **Bandwidth** (7.8–500 kHz), **Spreading** (SF5–12), **Coding rate** (4/5–4/8), **Path hash** (1–3 bytes per hop) and the airtime. Save asks **Apply radio change?**.

Scope 118

The default region for floods you send. Scope.

Quick replies

Edit, clear and reorder the ten presets.

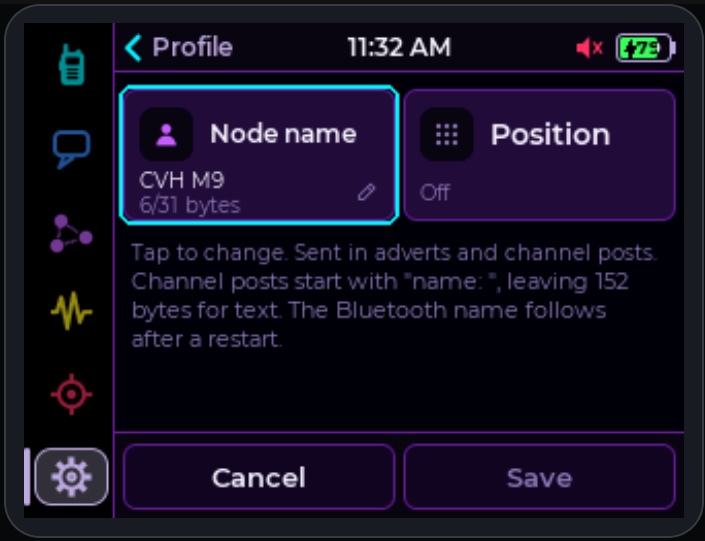
Auto retry ON

A tap flips it. **3 path · 2 flood** or **Off**. Details.

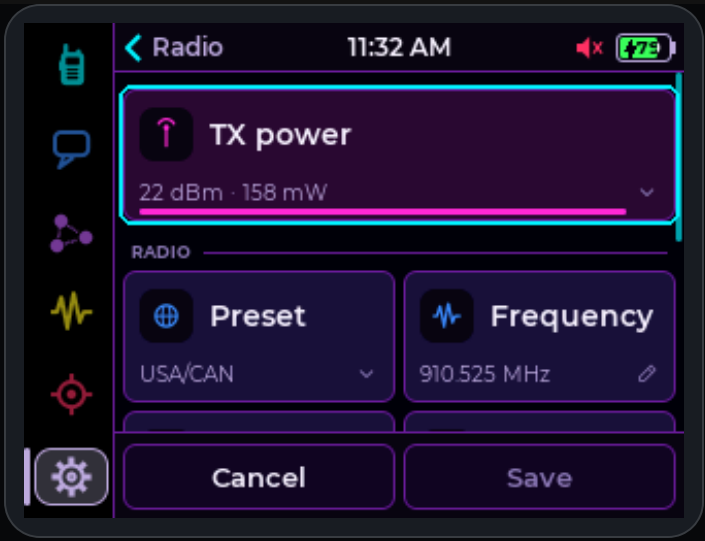
PRESET	FREQUENCY	BANDWIDTH	SF	CR
USA/CAN	910.525 MHz	62.5 kHz	7	5
EU/UK 869	869.618 MHz	62.5 kHz	8	5
EU classic	869.525 MHz	250 kHz	11	5

TX POWER

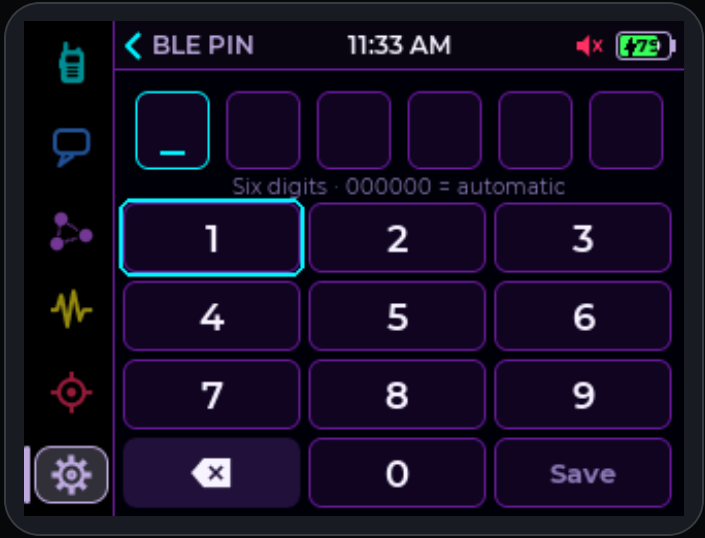
Keep transmit power within your region's limit and the board's rating. On the Heltec kit and the T096 the Radio pages show the power at the antenna (up to 28 dBm), and the setting stops at 18, where the amplifier is already at full output. The Heltec kit also has **FEM LNA** and **Boosted RX** switches under RECEIVE.



M9 Profile.



M9 Radio.



M9 Change PIN.

Connect

Wi-Fi

Wi-Fi, **Network**, **Net time**, **App link**, **Trusted**, **Saved**.

Bluetooth

Bluetooth, **App**, the pairing PIN, **Auto PIN**, **Change PIN**.

The Wi-Fi and Bluetooth scanners live on the Radios tab.

Hardware

Display

Brightness; **Key light** (M9, T-Deck); **Timeout** (5 s to Never); **Stay on USB**; **Msg wake**; **Text size**; **Compact** one-line messages; **Units**; **Battery saver** (M9, T-Deck, Heltec kit).

Sound

Tones (the mute switch), **Volume** on speaker boards, a sound per alert (Direct msg, Channel, Advert, Sent, Startup) and **Test sound**.

Buttons

Click, Double-click and Long press actions (**Space** and **Space x2** on the T-Deck). List.

Lock

Opens **Wake & lock**: **Tap to wake**, **Auto lock**, **Touch unlock**.

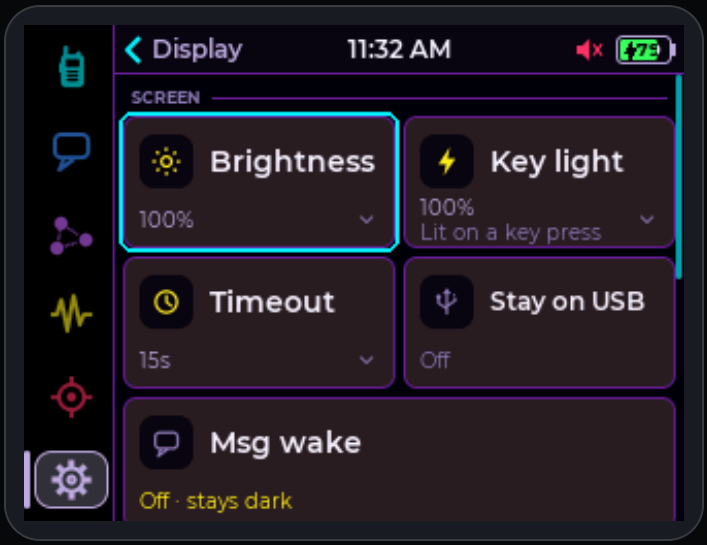
Storage

The card, USB drive, Files, Eject, Re-scan, Format.

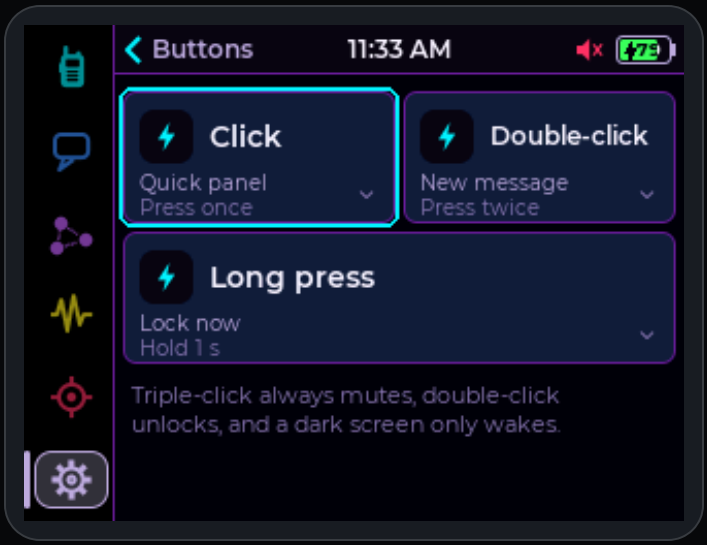
Hibernate

Power off until the User / PRG button. Hold 1.5 s, then confirm.

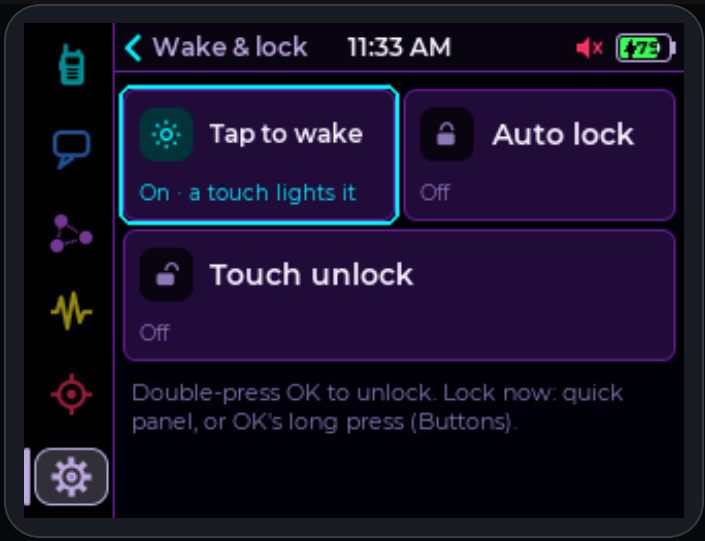
Built-in tones: None, Chirp, Kerplop, Tick, Advert, Ding, Double, Rising, Falling, Bell, Trill. Defaults: Direct msg Chirp, Channel Kerplop, Advert Advert, Sent Tick. Startup sounds: None, Chime, Soft, Classic. On the Heltec kit and M9 the piezo plays them as beeps, with no volume.



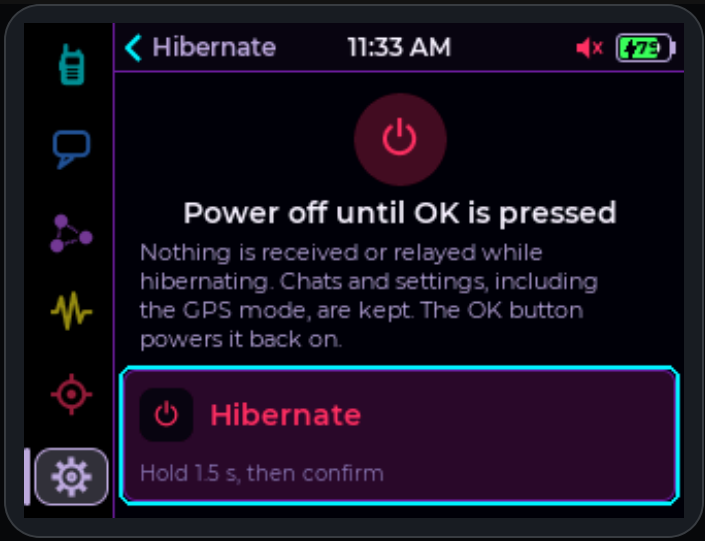
M9 Display.



M9 Buttons.



M9 Wake & lock.



M9 Hibernate.

Location

GPS

Mode (Off, On, Eco), **Receiver** (live state), **Last fix**.

Map

Which map set the map prefers, and a link to Storage.

Units

A tap flips Imperial and Metric (also in Display).

Appearance

Theme

Nine themes, each card drawn in its own colours. Previews live.

Screensaver

Style, brightness, what messages show, when it starts, and whether it runs on battery. **Preview** shows it now.

System

About

Build, node name, slot (**app0 · OTA**), MeshCore version, public key (tap for the QR), contacts n/350, channels n/40, health and the last restart.

Clock

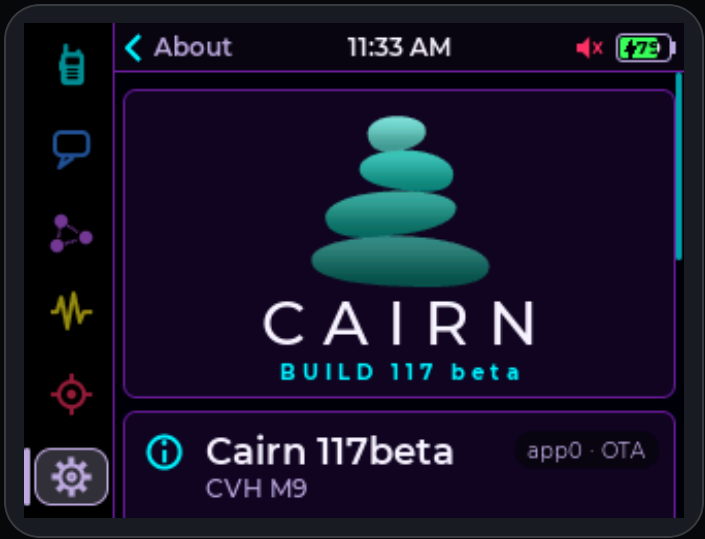
Style (12h, 24h, Analog), **Time zone** (US zones or Custom offset), **Daylight**, **Always sync**.

Firmware

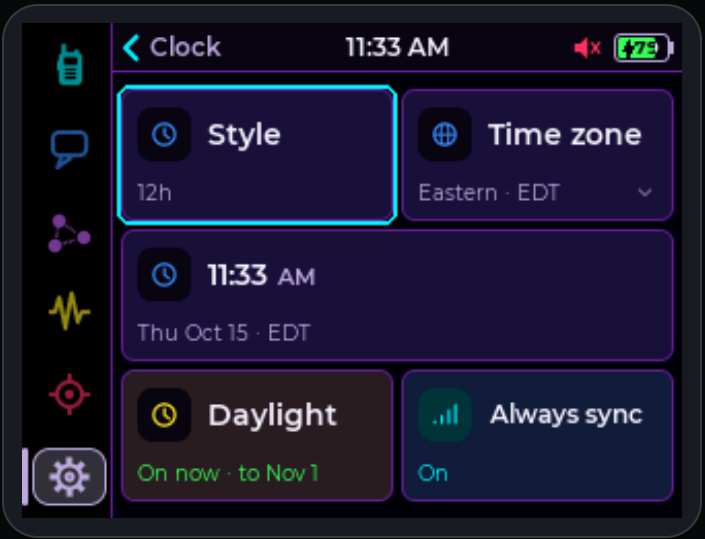
Opens **Firmware update**: the build, **Updates**, **Wi-Fi**, **Wi-Fi transfer**.

Restart

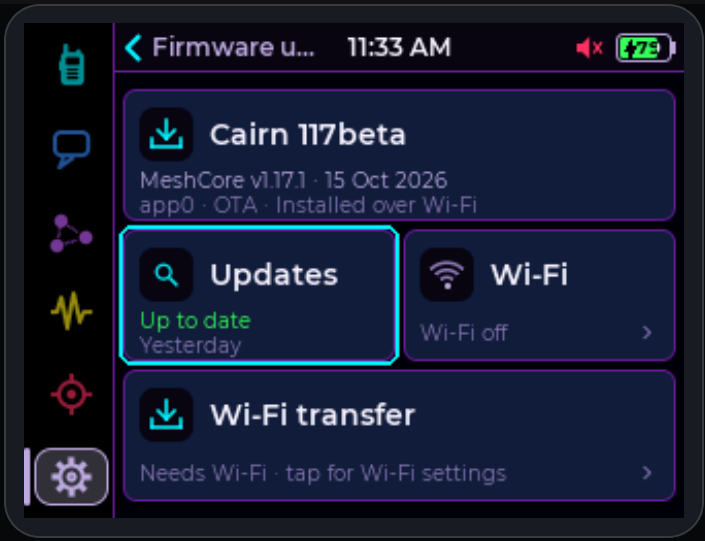
Opens **Maintenance**: last restart, **Restart**, **Reset UI**, **Factory reset** (hold 2 s), **Keep chats**, **Clear chats** (hold 1.5 s).



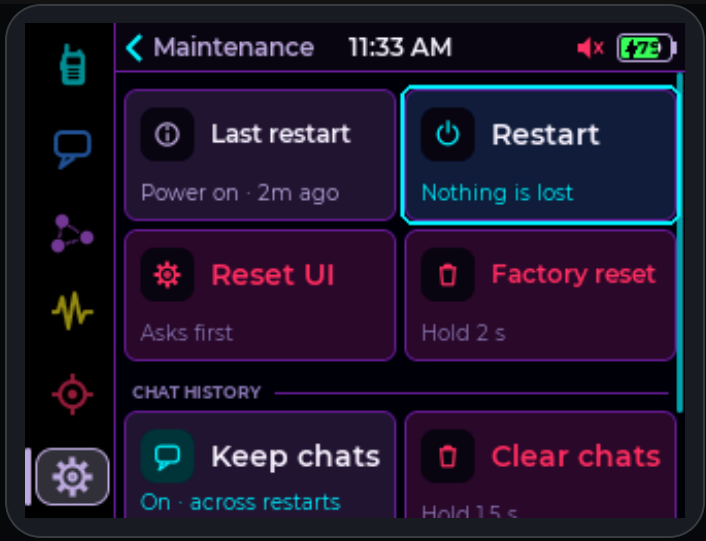
M9 About.



M9 Clock.



M9 Firmware update.



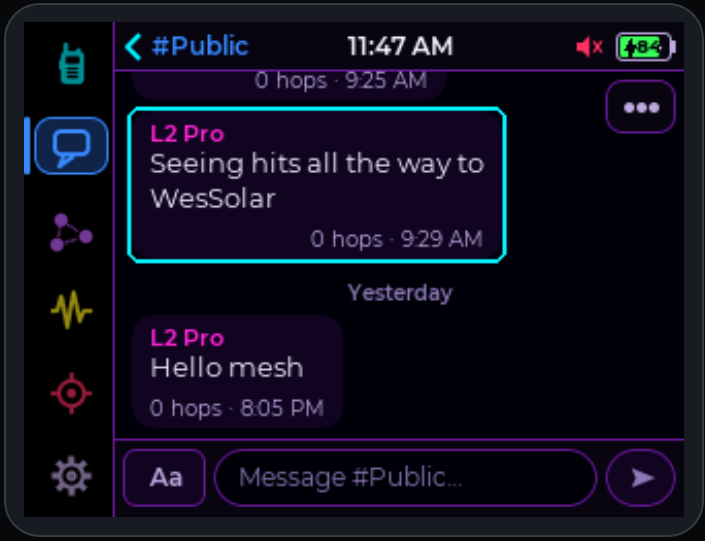
M9 Maintenance.

RESET

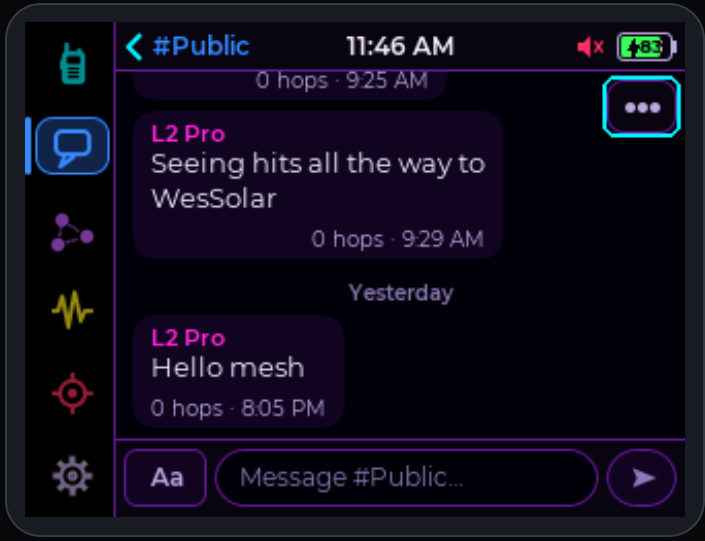
Reset UI resets units, tones, volume, buttons, lock touches and the map view; radio, contacts, chats and Wi-Fi are kept. **Factory reset** erases contacts, channels, chats, this node's identity, Wi-Fi and every setting, then restarts. Maps on the card are kept.

A ring does what a finger would.

The ThinkNode M9 has no touchscreen; the T-Deck has one and a keyboard with a trackball. Both drive the same interface by keys: a focus ring shows what is selected, the arrows move it, and `OK` taps it exactly as a finger would.



`M9` The ring outlines a message's bubble. `Hold OK` opens its card page.



`M9` The ring on ···, one `→` from the message field.

The keys

DOES	THINKNODE M9	T-DECK
Move the ring	d-pad `←` `→` `↑` `↓`	roll the trackball
Tap what is selected	`OK` (d-pad centre)	click the trackball
Long press what is selected	hold `OK`	hold the trackball in
Back	`BACK`	`⌫` outside a text field; hold the trackball with nothing selected
Home	`HOME`	the rail, or touch
Chats / new message	`MSG` / `@`	double `Space` for New message (default)
Send advert	`ADV` (hold: GPS on/off)	Radios tab, `OK` `OK`
GPS map	`MAP`; on the map: pan mode	GPS tab › Map
Quick panel	`CTRL` (again: close)	`Alt`+`C`, or `Space` (default)
Screen off	`MIC`	quick panel **Screen**
Recorder	—	`Mic`
Finish a text field	`OK` with nothing selected	keyboard `Enter` (Send, Done, Save), whatever the ring is on
Delete	`DEL`; hold: the word before the cursor	`⌫`
Radios tabs	the ring	`1` `2` `3`
Key light	Settings › Hardware › Display, or the quick panel slider	same, and `Alt`+`B` toggles it

How the ring behaves

- **Any key wakes a dark screen** and does nothing else. Locked, `OK` twice (or a hold) unlocks; on the T-Deck click the trackball twice.
- **Lists:** `↑` `↓` step through the rows; past the ends they leave the list. The M9's keyboard sends one press per key, so holding an arrow moves once. Quick presses go further: after six, each moves two rows, then four. A fast roll of the T-Deck's trackball accelerates the same way.
- **Sliders and steppers:** `←` `→` change them at once. `OK` enters adjust mode (amber ring): then `↑` `↓` change them too, and up always means more.
- **Hold-to-confirm tiles:** hold `OK`; a short press says **Hold OK to confirm**.
- **Pages with nothing to select** (Battery, About) show no ring: `↑` `↓` scroll, `←` goes to the rail, `BACK` leaves.
- **Focus is remembered** per page, so coming back lands where you were.
- **T-Deck:** touching the screen hides the ring; the next arrow brings it back where it was. Characters, Delete and the function keys act whether the ring shows or not.

Typing and chats

- In a thread, **type a letter**: the text box opens with it. In the new-message picker, typing searches.
- `→` from the message field (or Send) jumps to the chat's ⋯, instead of stepping up through the whole history.
- On a message, the ring outlines its bubble; `OK` opens the Reader and `Hold OK` the card page (Reply, Copy, Resend…).
- The emoji and ?123 panels, the Paste chip and Send are all reachable with the arrows.
- T-Deck: `⌫` on an empty field leaves it, but a run of deletes that empties the field never leaves by accident. Holding the trackball leaves without erasing.

> **TIP**
> On the Battery page, `OK` on the graph puts a cursor at now; `←` `→` move it back through time, with bigger steps as you keep pressing, and `BACK` clears it.

NOT BY KEYS

Not reachable by keys yet: the Home header's long press (use **Settings › System › Clock**), the Spectrum plots' tap-for-readout, and the map's long-press menu. **Settings › Hardware › Buttons** configures a button the M9 doesn't have, and Tap to wake does nothing there.

Small screens, page by page.

The Wio Tracker L1 Pro family and the Heltec T096 run a compact Cairn interface. Pair the MeshCore app over Bluetooth for full messaging; the device shows messages, nodes, the radio, GPS and settings.

Wio Tracker L1 Pro, DV1, L1 e-ink

A five-way joystick and a Back button. Pages, in order: **Clock** (with a battery view), **Messages**, **Nodes**, **Radio**, **GPS** and **Settings**. Icons for the pages run along the bottom, or down the side with **Page tabs** set to Side.

BOTTOM TABS (DEFAULT)

- ← → change page.
- ↑ ↓ switch the page's own view (Clock / battery, the Radio and GPS views, Recent / Contacts) or move in a list.
- Press the joystick to select or open; Back steps out.

SIDE TABS

- ↑ ↓ step between pages; past Settings wraps to the first.
- ← → switch the page's own view.
- On Messages and Settings, press (or →) to step into the list; Back steps out.

- **Messages:** read threads; reply or compose by picking a recipient, then a preset. Messages sent from the phone app appear as yours, and a delivered direct message shows a ✓; channels keep a **>** marker.
- **GPS:** a joystick click steps through position, compass and GPS data (satellites, HDOP, grid square, fix age). **NO DATA** means the receiver has sent nothing for 10 s.
- **Settings** opens on six categories: Radio, Connect, Location, Display, Sound and System. Inside a category, ↑ ↓ pick a row and press (or →) edits it; in an editor ← → change the value. ← or Back returns to the categories. System › About shows the build, MeshCore version and key prefix. Path hash and Units are here too.
- A thin line under the title flashes when the radio sends (dashed) or receives (solid). Not on e-ink.

IMAGES

Pick the image for your screen: `WioL1Pro` (SH1106 OLED, most units), `DV1` (2.42" SSD1309 on the Grove port) or `WioL1Eink` (2.13" black-and-white e-ink; three-colour panels aren't supported). `WioL1RoomServer` turns the L1 into a room server.

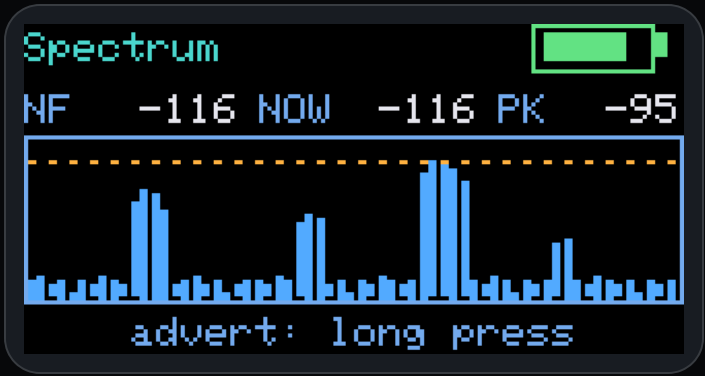
Heltec Mesh Node T096



T096 · SIMULATED Home: unread count and the pairing PIN.



T096 · SIMULATED Recently heard.



T096 · SIMULATED Spectrum: hold to send an advert.

One button. **Click**: next page. **Hold**: the page's action, as the bottom line says (for example, send an advert on Spectrum). Reboot and Hibernate arm on the first hold and act on a second. Holding the button during the first 8 s after boot opens the rescue console. Settings is one list: Theme (Dark, Color, Night, Soft, Stock), Screen, Keep on USB, Bluetooth, GPS, Reboot, Hibernate, Back. The `-ble` image is the main build; `-usb` links the app over USB serial instead (no Bluetooth, no GPS page).

A square screen and a keypad.

The LilyGO T-Display SF32 runs its own card interface on a 480×480 AMOLED, by touch or by the 20-key keypad board. It has Home cards, Chats, Nodes with node pages, Radios with Traffic, GPS, an IR remote, sensors, and a player and recorder.

CONTROL	DOES
▲ / ▼	move up and down; on values and sliders, up always means more
OK	select; hold: context menu
Back	back; on Home with the sidebar selected, screen off
Home	Home tab; hold: quick panel
Mail	newest unread message; hold: new message
Stop	stop an alert or dismiss a pop-up; hold: mute
0 – 9	Home card shortcuts; letter jump in lists; typing in chats (multi-tap: 2 abc … 9 wxyz, 1 punctuation, 0 space; # cycles Abc / abc / ABC / 123; * symbols and emoji)
Radios	1 LoRa, 2 Wi-Fi, 3 BLE; scanners: 4 Active, 5 Passive, 6 Plot / List, 7 Start
Side C	click: screen off or on; double: unlock; hold: lock
Side A	click: Settings; double: new message; hold 1–3 s: back; triple: mute; hold 3 s: power menu (Hibernate, Restart); wakes from hibernate; hold 12 s: hard restart
Side B	click: alert profile; hold: flashlight
Side D	hold 2 s: wake after a low-battery power-off; hold about 20 s: power cut and restart

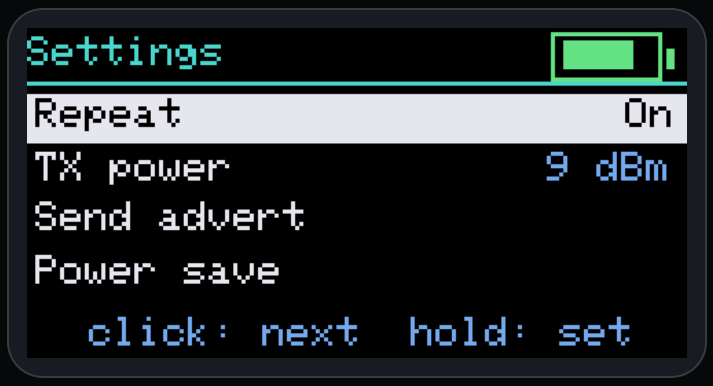
By touch: swipe right from away from the left edge to go back. The keyboard board carries the Wi-Fi chip, so Wi-Fi, online updates and Wi-Fi scan need the core docked; while Wi-Fi is on, the GPS pauses. Docked and on battery, sliding the power switch off shows a 5-second **Switched off** card (any key cancels), then saves and turns off. No offline map yet.

A repeater you can read at a glance.

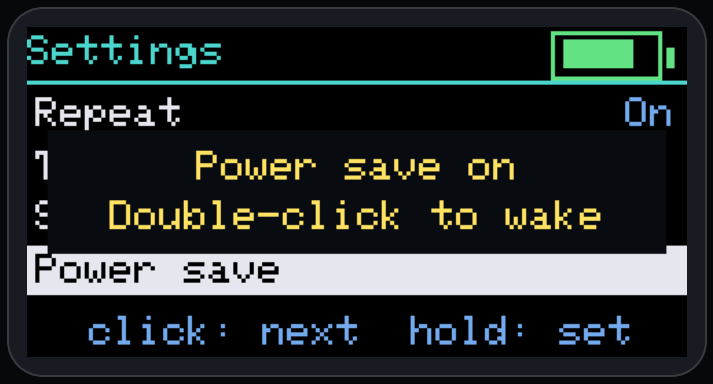
The compact repeater is the standard MeshCore repeater (same mesh behaviour, CLI and remote admin) with a minimal Cairn screen on one button.



T096 · SIMULATED Status.



T096 · SIMULATED Settings.



T096 · SIMULATED Power save.

PAGE	SHOWS	HOLD
Status	repeat on/off, the channel, packets sent and heard, neighbours, noise floor, uptime, battery	turn off: keep holding about 3 s until **Release to turn off**
Spectrum	the channel over the last ~16 s, with floor, latest and peak	send an advert
Settings	Repeat, TX power, Send advert, Power save, Reboot	change or run the selected row; Reboot asks for a second hold within 5 s

Click moves to the next page (on Settings, the next row). The screen turns off after 30 s (2 minutes on Spectrum); the first press only wakes it. **Power save** turns the screen and the TX

LED off and is kept across reboots; **double-click** wakes it. After turning off, a press turns the repeater back on.

A board coming from a companion image starts as a new repeater with the admin password `password`. Set it up over USB serial (115200 baud), for example for the US:

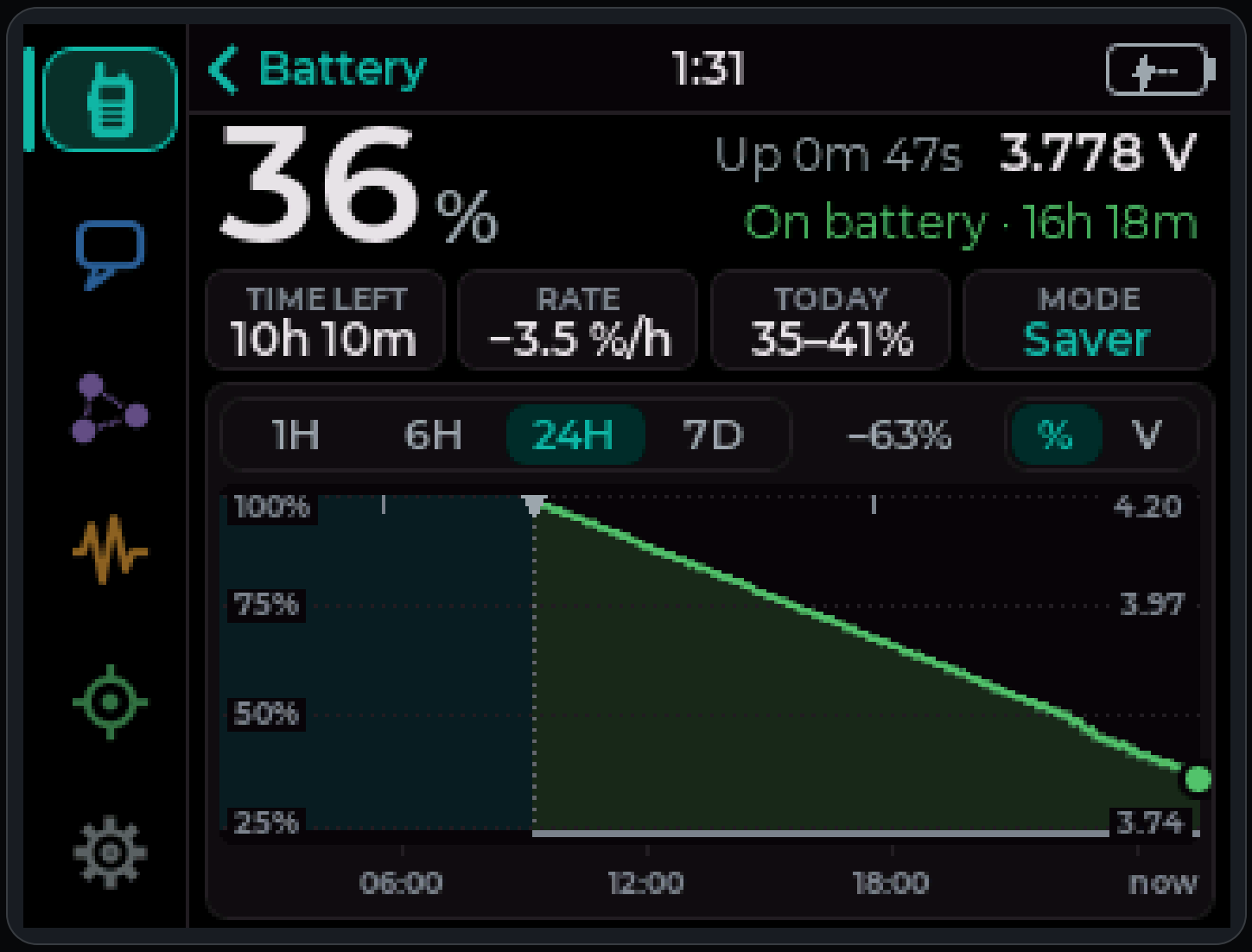
```
set name Ridgetop Repeater
set radio 910.525,62.5,7,5
set tx 9
password MyAdminPass
set lat 35.1234
set lon -97.5678
reboot
```

TX power steps through 1, 5, 9, 13, 17 and 22 dBm of radio-chip output; the T096 and Heltec V4 add their amplifier's gain on top. Repeater regions and the rest of the CLI are standard MeshCore.

Make the battery last.

The Battery page

Tap Home's Battery card. The header shows the percent, the voltage and a status line (**On battery · 16h 18m**, **USB · charging**). Cells show time left or to full, the rate, today's low and high, the last full charge and the mode. The graph covers **1H**, **6H**, **24H** or **7D** in % or volts, with markers for plug-in, unplug, full, warnings and restarts. Tap the graph for a readout of that moment. The week's history comes back after a restart, once the clock is set.



T-DECK · 113 The Battery page.

- **Charging** and **Full** are estimated from the voltage on most boards, so the percent reads high while charging. On the T-Deck the cell can't be read on USB: it says **USB · charging**;

the blue LED going out means charged. On the M9 the reading pauses while Wi-Fi is on (**Reading paused · Wi-Fi**).

- **Low battery** warns at 20 % and 10 % with a tone, then the device turns itself off before the cell is drained. Plugging in cancels it.
- **Battery saver** (**Settings › Hardware › Display**, off by default, on the M9, T-Deck and Heltec kit): while the screen is dark the chip sleeps between radio packets. On the T-Deck it was measured at about 43 mA instead of 82 mA with the screen off, about 54 h of standby instead of 28 h, and it still received every packet. On the M9 it also turns Bluetooth off, and turning Bluetooth back on restarts the device.
- **GPS** is the biggest single drain. Use Eco or Off when you don't need tracking.

> TIP
> **Stay on USB** keeps a lit screen on while plugged in, and the screensaver then takes over. On the Heltec kit this works on a computer's port only, since the board can't sense a wall charger.

Hibernate

Settings › Hardware › Hibernate: hold the tile 1.5 s, then confirm. The device receives and relays nothing until you press `User` (L2 Pro) or `PRG` (Heltec); WAKE / IO can't wake it. Settings and chats are kept (chats only with **Keep chats** on). On the M9, use the power slider (off, then on) or RESET to wake it. Prefer Hibernate to a hardware power switch: it saves first.

When something is off.

SYMPTOM	WHAT TO DO
Messages typed on the device stop at the first repeater, phone messages go through	Your mesh uses regions. Set **Settings › Radio › Scope** to the same region the app uses, or set a channel's own Scope.
A direct message says **not delivered**	Hold it › **Resend**. If it keeps failing, ⋯ › **Reset path** (or **Find path**) and send again. Consider Auto retry.
The clock reads --:--	Normal after a power cut. Get a GPS fix, connect the app, or join Wi-Fi with Net time on.
Repeater login is greyed out	**Clock not set · logins will be ignored**: set the clock first.
The phone can't pair	Read the current PIN on Home's Network card; it changes each boot unless fixed. Check Bluetooth is on and Wi-Fi transfer is not running.
The app over Wi-Fi is refused	Answer **Allow app over Wi-Fi?** on the device. A refused address is blocked for 2 minutes.
Map shows **Card isn't FAT32** or **No maps on card**	Format the card FAT32 (on the device or with the computer) and build sets into `/Maps` with `tools/maptiles.py`.
A song is refused: **Too fast for this card: use MP3**	T-Deck: convert large WAV files to MP3.
A USB update "worked" but the build didn't change	The board runs from its second slot. Run `esptool --chip esp32s3 erase-region 0xe000 0x2000`, then write the `-update.bin` again.
Heltec V4 boot-loops after install	The touch build needs a V4-R8 (`esptool flash-id` reports *Embedded PSRAM 8MB*) in the Expansion Kit V2. A standard V4 needs the HeltecV4 preview build.
UF2 copy fails on a Mac with error -36	`cp -X file.uf2 /Volumes/<drive>/` from Terminal.
SF32 installer can't find the board or stops	Power switch ON with a charged battery; a data cable straight into the computer; close serial monitors. Reset: hold A 12 s, then run the installer again.

SYMPTOM	WHAT TO DO
A beta isn't offered by Firmware update	By design: betas install by hand. The next full release will be.
Last restart shows **Crash**, **Watchdog** or **Brownout** in amber	The device restarted after a fault. Note the build number and report it.
The device seems stuck	L2 Pro / Heltec: tap RST. M9: RESET. T-Deck: power switch off and on. SF32: hold A 12 s.

REPORT

Bugs: github.com/cvhviz/Cairn/issues. Include the board, the build number from the splash, and what you did.

Questions people ask.

Does Cairn work with the MeshCore apps?

Yes. Every build is MeshCore v1.17.1 underneath and works with the official apps over Bluetooth, and on the L2 Pro, Heltec kit, M9 and T-Deck over Wi-Fi too.

Do I lose my contacts when I update?

No. Updates (online, Wi-Fi transfer, `-update.bin`, UF2, the SF32 installer) keep the identity, contacts, channels and settings. Only a first ESP32 install with the `-full.bin` after a full erase starts over.

Do messages I send from the phone show on the device?

Yes, as yours, with the delivered tick when confirmed. With CairnOS, messages sent on the device also show in the app.

Why two ticks on direct messages but not on channels?

A direct message is acknowledged by its recipient. A channel has no single recipient, so it shows **heard N**, how many repeats your radio heard.

How many messages does it keep?

200 in total, across every conversation. They survive restarts while **Keep chats** is on, and are stored unencrypted.

Will Auto retry flood the mesh?

It is designed not to: three tries by the known path first, at most two floods, with growing waits over about two and a half minutes. It is off by default.

Can I set a custom frequency?

Yes, in
Settings › Radio › Radio
: frequency, bandwidth, spreading factor and coding rate are all editable. Every node on a mesh must match.

Does the map download tiles?

No. The device never downloads maps; build sets with `tools/maptiles.py` and copy them to the card.

Which boards play music?

The L2 Pro and the T-Deck (and the T-Display SF32, in its own interface). The Heltec kit and M9 have only a buzzer.

The words on the screen.

Advert

A short broadcast that announces your node: its name, its key and, if you allow it, its position. **Zero-hop** reaches nodes in direct range; **Flood** is relayed across the mesh.

Flood

A packet every repeater passes on, used when no route is known. Floods cost airtime across the whole mesh.

Direct path

A stored route to a contact through specific repeaters. Messages with a path go only that way. **Direct** or **D** means no hops at all.

Hops

How many repeaters relayed a packet. **0 hops** means heard directly.

Path hash

How many bytes identify each hop in a flood's path: 1, 2 or 3. Larger is less ambiguous on big meshes.

Scope / region

A label on a flood that regional repeaters check. A repeater set to regions only passes floods scoped to a region it allows.

Repeater

A node that relays other nodes' packets. You can scan for repeaters and log in to administer them.

Room server

A node that hosts a shared message room. Log in to read and post; it appears in Chats as **@Room**.

Companion

A node that people message from, usually paired with a phone. Cairn on your device is a companion.

ACK

The acknowledgement a recipient sends back for a direct message. It turns one tick into two.

Channel

A group conversation readable by everyone with its key. **Public**: the well-known key. **Hashtag**: the key comes from the name. **Private**: a random key shared by hand.

SNR / RSSI

Signal-to-noise ratio in dB (higher is better) and received signal strength in dBm.

Noise floor

The background radio level in dBm. **MARGIN** is how far a signal stands above it.

Airtime

How long a transmission holds the channel. Slower presets cost more airtime per message.

Heard N

On your own message: how many repeats of it your radio heard.

Buttons, ports, features.

Wio Tracker L2 Pro

ESP32-S3 · SX1262 · 3.2" TOUCH

USER	click quick panel, double new message, hold back, triple mute; wakes from hibernate
WAKE	click screen off/on, double unlock, hold lock
FLASH	hold User, tap RST, release
HAS	GPS, microSD, speaker and mic, Wi-Fi, USB drive

Heltec V4-R8 + Expansion Kit V2

ESP32-S3 · SX1262 + FEM · 2.8" TOUCH

PRG	as the L2's User; the only button that wakes from hibernate
IO	as the L2's WAKE
FLASH	hold PRG, tap RST, release; never hold IO with it
HAS	GPS, microSD, piezo, Wi-Fi, Battery saver, FEM LNA and Boosted RX

Elecrow ThinkNode M9

ESP32-S3 · LR1110 · 2.4" KEYS BETA

KEYS	d-pad and OK, BACK, HOME, MSG, @, ADV, MAP, CTRL, MIC, DEL
POWER	slider disconnects the battery; RESET restarts
FLASH	automatic over USB-serial
HAS	GPS, RTC, microSD, piezo, Wi-Fi, key light; offline map untried

LilyGO T-Deck / Plus

ESP32-S3 · SX1262 · 2.8" TOUCH + KEYS BETA

TRACKBALL	moves the ring, click is OK, hold is long press
KEYBOARD	Space configurable, Alt+C quick panel, Alt+B key light, Mic recorder, Enter finishes a field
FLASH	switch off, hold trackball, switch on
HAS	microSD, speaker and mic, Wi-Fi, USB drive; GPS on the Plus

LilyGO T-Display SF32

SIFLI SF32LB52 · SX1262 · 480×480 AMOLED

SIDE	A, B, C, D (see chapter 18)
KEYPAD	▲ ▼ OK Back Home Mail Stop, 0–9 # *
INSTALL	the -install.zip installer
HAS	GPS, Wi-Fi when docked, IR, sensors, audio; no offline map yet

Wio Tracker L1 Pro

NRF52840 · SX1262 · OLED OR E-INK

JOYSTICK	pages, views, select
BACK	back out of a page or menu
INSTALL	double-tap Reset, copy the .uf2
HAS	GPS, buzzer; room server image

Heltec Mesh Node T096

NRF52840 · SX1262 + PA · 0.96" TFT

BUTTON	click next page, hold action
RST	restart; double-tap for UF2
HAS	GPS, five themes, BLE or USB companion, repeater image

Feature matrix

FEATURE	L2 PRO	HELTEC KIT	M9	T-DECK	SF32	L
Input	touch	touch	keys	touch, keys	touch, keypad	jo
MeshCore apps over Bluetooth	✓	✓	✓	✓	✓	
Wi-Fi and online updates	✓	✓	✓	✓	✓ **docked**	
Scope pages, resend, auto retry, copy/paste (118)	✓	✓	✓	✓	default scope	defa
GPS	✓	✓	✓	Plus	✓	
Offline map	✓	✓	untried	✓	coming	
Spectrum and band scan	✓	✓	✓	✓	✓	
Wi-Fi and Bluetooth scanners	✓	✓	✓	✓	✓	
Repeater scan and admin	✓	✓	✓	✓	✓	
Music player, recorder	✓	–	–	✓	✓	
Alert sounds from the card (118)	✓	–	–	✓	–	
File manager	✓	✓	✓	✓	–	
USB drive	✓	–	–	✓	–	
Contact QR code	✓	✓	✓	✓	–	
Colour themes	9	9	9	9	9	

Who made this.

Cairn firmware, its interfaces and this manual: Christopher Van Hoose, CVH Visuals (cvhvisuals.com). Releases and source: github.com/cvhviz/Cairn.

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Cairn is an independent project and is not affiliated with Seeed Studio, Heltec, Elecrow, LilyGO or SiFli. Board names are their makers' trademarks. Screens in this manual were captured on devices and in the firmware's simulator; some show test data or fictional sample data, and some come from earlier builds where noted.